

Received 9 July 2025, accepted 17 August 2025, date of publication 20 August 2025, date of current version 9 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3600622

## RESEARCH ARTICLE

# Hybrid Framework Combining Diffusion-Based Image Augmentation and Feature Level SMOTE for Addressing Extreme Class Imbalance

RAJ KUMAR<sup>1</sup>, YONG-WOON KIM<sup>2</sup>, AND YUNG-CHEOL BYUN<sup>3</sup>

<sup>1</sup>Department of Electronic Engineering, Institute of Information Science and Technology, Jeju National University, Jeju-si 63243, South Korea

<sup>2</sup>Department of Computer Engineering, Institute of Information Science and Technology, Jeju National University, Jeju-si 63243, South Korea

<sup>3</sup>Department of Computer Engineering, Major of Electronic Engineering, Institute of Information Science and Technology, Jeju National University, Jeju-si 63243, South Korea

Corresponding author: Yung-Cheol Byun (ycb@jejunu.ac.kr)

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Ministry of Science and ICT, Korea government (MSIT) (No. RS-2024-00405278), and this work was also supported by the National Research Foundation (NRF), Korea, under project BK21 FOUR.

**ABSTRACT** Imbalanced datasets pose significant challenges to the reliability and robustness of visual classification systems, particularly in critical applications like solar panel dust detection. Conventional data augmentation techniques, such as Synthetic Minority Over-Sampling Technique (SMOTE) or generative models like Stable Diffusion, independently demonstrate limitations, either by lacking structural diversity or failing to sufficiently address feature-space imbalance. To overcome these constraints, this paper proposes DiSMHA (Diffusion-SMOTE Hybrid Augmentation), an innovative framework integrating pixel-level image synthesis using Stable Diffusion and feature-level oversampling via SMOTE. DiSMHA systematically balances and enriches minority class representation by producing high-quality synthetic images coupled with robust feature interpolation in the latent space. Experiments conducted on four imbalanced solar panel image dataset cases demonstrate substantial improvements across multiple performance metrics, particularly accuracy, recall and F1-score, emphasizing effective minority class detection. The proposed DiSMHA approach was evaluated using several classification models, including Support Vector Machine (SVM), Multilayer Perceptron (MLP), XGBoost, Vision Transformer (ViT-Head), and TabNet. These models achieved notable accuracies ranging from 94% to 98%, significantly outperforming baseline methods. Particularly, the MLP (with accuracy 96.67%) and SVM (with accuracy 94.67%) models exhibited substantial performance gains, confirming DiSMHA's capacity to enhance traditionally imbalance-sensitive classifiers. Comparative analyses further underscored the superiority of DiSMHA, highlighting its consistent accuracy, enhanced recall, and improved F1-scores over individual augmentation approaches. Moreover, statistical validation using the Wilcoxon signed rank test confirmed the significant superiority of our approach for minority class detection in highly imbalanced scenarios. Thus, DiSMHA represents a scalable, robust solution applicable across diverse domains facing imbalanced classification challenges, including industrial inspection, medical imaging, and rare event detection.

**INDEX TERMS** Hybrid data augmentation, imbalanced classification, SMOTE, stable diffusion, solar panel fault detection, convolutional neural networks, DiSMHA, feature-level oversampling.

## I. INTRODUCTION

Deep learning models have revolutionized various visual recognition tasks, from object detection to the identification

The associate editor coordinating the review of this manuscript and approving it for publication was Li He<sup>1</sup>.

of rare exceptions [1]. However, real-world image classification problems often suffer from a critical challenge: imbalanced datasets, where important classes are vastly outnumbered by other classes. This imbalance leads to biased models that fail to detect rare yet essential instances in practical applications [2]. As a result, recall of the minority

class is compromised, which results in high misclassifications for important but low events or conditions. This problem is not unique to a domain; There is a wide range of real-world scenarios. For example, in industrial visual inspection, defective products are much less common than non-defective products, but it is still extremely important to detect rare defective cases [3]. Similarly, in medical imaging, the detection of rare diseases in a large population of normal samples is important for effective diagnosis and timely intervention [4]. In such cases, standard deep learning methods perform poorly, as the model is dominated by the majority class and thus fails to learn a strong representation for the minority class [5]. Therefore, there is a need for an innovative and effective framework that can deal with this problem by making datasets more robust and balanced. Traditional solutions, like oversampling or basic data augmentation, often fall short, either by creating unrealistic samples or failing to enrich the underlying data diversity. To address this challenge, we present a hybrid data augmentation approach (shown in Fig. 2) that blends the power of advanced image generation and feature-level techniques, aiming to significantly boost minority class recognition and deliver more reliable, robust classification results.

Although several methods have been proposed to address the issue of class imbalance, many suffer from inherent limitations. Traditional oversampling techniques, such as the Synthetic Minority Oversampling Technique (SMOTE), interpolate between existing minority class feature vectors in order to generate new synthetic samples in the feature space [6]. While this approach effectively increases the representation of minority classes at the feature level, it does not introduce new structural diversity at the pixel level. This limitation often restricts the model's ability to capture the true variability within minority classes, leading to suboptimal generalization. Conversely, image-space augmentation methods—such as those based on Generative Adversarial Networks (GANs) or, more recently, diffusion models like Stable Diffusion—generate novel images from textual prompts or noise [7]. These models have demonstrated an impressive ability to produce realistic and diverse samples, thereby enhancing the pixel-level diversity of the minority class. However, existing augmentation strategies typically operate in isolation: feature-space oversampling methods like SMOTE do not leverage the pixel-level information, while generative models focus solely on the image space and often lack control over feature-level diversity and class separation [8]. As a result, neither approach alone suffices to address the dual challenge of capturing fine-grained intra-class diversity and ensuring robust feature representation for minority classes. This research presents a critical gap in the current literature and motivates the need for a unique hybrid framework that leverages the strengths of both pixel-level and feature-level augmentation.

The significance of class imbalance extends far beyond academic interest, influencing the deployment of AI models

in high-stakes environments such as healthcare, security, and industrial automation [9]. In these domains, the cost of misclassifying minority classes can be substantial—missed diagnoses, undetected defects, or overlooked security threats can result in severe financial, social, or health-related consequences. Therefore, any advancement that can improve the performance of minority classes holds immense practical value. A hybrid augmentation approach that combines the strengths of both state-of-the-art generative models and classical feature-level oversampling not only enhances the training data's diversity but also paves the way for robust, scalable solutions applicable across domains [10]. The integration of Stable Diffusion—a powerful diffusion-based generative model—with SMOTE can provide both structural varieties at the pixel level and greater dispersion in the latent feature space, thereby overcoming the individual limitations of each technique.

This research aims to bridge the existing gap by proposing a novel, hybrid data augmentation framework (DiSMHA) that synergistically combines diffusion-based image generation with feature-level oversampling. Through this dual augmentation approach, the study seeks to significantly improve classifier robustness, especially for minority classes, and to provide a scalable template for addressing class imbalance in diverse applications. The key contributions of this research are as follows:

- We propose a novel hybrid augmentation framework, Diffusion-SMOTE Hybrid Augmentation (DiSMHA), that combines Stable Diffusion-based synthetic image generation with feature-level SMOTE oversampling to address class imbalance in visual classification.
- We systematically evaluate the effectiveness of DiSMHA across multiple imbalance scenarios using a real-world solar panel dust dataset, demonstrating consistent improvements in minority class recall, F1-score, and overall accuracy.
- We conduct extensive comparative experiments by using five classification models (SVM, MLP, XGBoost, ViT-Head, TabNet), showing that DiSMHA outperforms both non-augmented and single-method augmented baselines.
- We provide practical insights and comprehensive analysis on the impact of hybrid augmentation in imbalanced scenarios, including the discussion of real-world applicability to other domains.

This paper is organized into several sections to comprehensively present the proposed hybrid data augmentation approach to mitigate data imbalance. Section II reviews the literature on class imbalance, data augmentation, and the use of diffusion models and oversampling techniques. Section III presents the detailed methodology of the proposed hybrid augmentation pipeline, including data preprocessing, image generation, feature extraction, and SMOTE application. Section IV reports and analyzes the experimental results, with

comparisons to baseline and alternative methods. Section V discusses the findings, implications, and limitations of the study. Finally, Section VI concludes the paper and suggests directions for future work.

## II. RELATED WORK

Class imbalance poses significant challenges in machine learning, particularly affecting visual classification tasks where minority classes are critically underrepresented. Traditional techniques such as oversampling (SMOTE, ADASYN) and undersampling have been widely utilized but exhibit limitations when applied in isolation. In recent years, generative models, including GANs, VAEs, and diffusion models, have emerged as powerful tools for synthetic data generation, addressing some limitations of traditional methods. However, standalone generative approaches also possess drawbacks, prompting research into hybrid methods combining classical and generative approaches, shown in Table 1. This literature review systematically explores classical oversampling/undersampling, generative augmentation methods, hybrid approaches, and their applications, highlighting existing gaps and motivating the development of DiSMHA, a hybrid pixel-feature augmentation.

### A. CLASSICAL OVERSAMPLING OR UNDERSAMPLING

Classical data balancing techniques such as oversampling and undersampling are foundational in addressing class imbalance. Mohammed et al. [11] conducted an extensive comparative study employing oversampling methods, including SMOTE, random oversampling, and undersampling strategies, using datasets like the Santander Customer Transaction Prediction from Kaggle. They observed oversampling significantly improved minority class detection but noted limitations such as overfitting and reduced generalization when applied excessively. Furthermore, Wongvorachan et al. [12] specifically investigated the performance of SMOTE-NC and traditional sampling methods within educational data mining contexts using the High School Longitudinal Study of 2009 dataset. Their empirical results demonstrated superior performance of SMOTE-NC in handling categorical and numerical data, emphasizing SMOTE's adaptability but pointing out the limitations in image-based applications. Similarly, Tyagi et al. [13] provided comprehensive insights through the analysis of 15 real-life datasets, illustrating that oversampling generally outperformed undersampling across diverse classifiers. However, they emphasized oversampling's tendency towards overfitting and undersampling's potential information loss, further underscoring the need for integrative methods. Further exploring ensemble methodologies, Ding et al. [14] introduced KA-Ensemble, combining Kernel-ADASYN with ensemble learning approaches to tackle imbalanced image classification tasks. Their innovative approach yielded notable improvements in classification accuracy and robustness compared to standalone methods. Moreover, Zhang and Chen [15] implemented clustering-based undersampling combined with random

oversampling to classify breast cancer diagnosis datasets. Their findings affirmed enhanced classification performance but indicated limited applicability beyond the medical domain, highlighting the necessity for broader, generalized methods. Dey and Pratap [16] rigorously evaluated SMOTE, Borderline-SMOTE, and ADASYN using well-established datasets such as Pima Indian Diabetes, Breast Cancer Wisconsin, and Insurance Claim Prediction datasets from Kaggle. They observed Borderline-SMOTE exhibiting superior performance due to better handling of boundary samples, yet noted a clear absence of feature-level diversity. Masruriyah et al. [17] extended this evaluation by analyzing heart disease datasets using SMOTE and ADASYN. Their results underscored SMOTE's consistent performance improvement but also pointed out feature-level homogeneity issues.

### B. GENERATIVE APPROACHES FOR AUGMENTATION

Generative approaches, including GANs, VAEs, and Diffusion Models, have revolutionized data augmentation due to their ability to produce realistic and diverse synthetic data. Lin et al. [18] further elaborated on these aspects, demonstrating how GANs, VAEs, and diffusion models effectively augmented semiconductor manufacturing datasets, significantly improving robustness. Yet, they indicated limitations in broader generalization, suggesting further research in general visual classification tasks. Specifically, diffusion models have gained significant attention due to their advanced image synthesis capabilities. Ayton and Nelson [19] examined diffusion models' effectiveness for augmenting medical imaging data, reporting substantial improvements in classifier performance. However, they acknowledged the need for comprehensive evaluations across varied image domains to validate diffusion models' universal applicability. Furthermore, Vivekananthan [25] comparatively analyzed VAEs, GANs, and Stable Diffusion, highlighting diffusion models' superior balance of realism and diversity. Yet, they identified a critical gap: these models were not specifically designed for class imbalance mitigation. Further reinforcing this notion, Kumar [26] conducted a detailed review of diffusion models, outlining their broad applicability but highlighting their unexplored potential in handling class imbalance directly. Rais et al. [20] compared Disc-VAE and GAN for medical image generation, noting GAN's superior realism and Disc-VAE's improved feature space representation. Moreover, Kanyepe et al. [27] explored generative models broadly across multiple modalities, concluding with recommendations for detailed, domain-specific investigations. Furthermore, Suresh and D'Ettorre [21] demonstrated diffusion models' effectiveness in medical imaging but emphasized limited exploration in other significant domains such as defect detection or rare event classification.

### C. HYBRID APPROACHES AND THEIR LIMITATIONS

Recent research has shifted toward hybrid approaches to leverage the strengths of both classical and generative

**TABLE 1. Previous studies on multiple augmentation techniques to balance the dataset.**

| Ref  | Year | Dataset                                                                                | Augmentation Technique                                  | Strengths                                                                                                         | Limitations                                                                      |
|------|------|----------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| [11] | 2020 | Santander Customer Transaction Prediction (Kaggle)                                     | SMOTE, Random Oversampling, Random Undersampling        | Provided a comparative study on various sampling techniques, highlighting their impact on classifier performance. | Lacked focus on image data and did not explore generative augmentation methods.  |
| [12] | 2023 | High School Longitudinal Study                                                         | SMOTE-NC, Random Oversampling, Random Undersampling     | Demonstrated the effectiveness of SMOTE-NC in educational data mining.                                            | Focused on tabular data; not applicable to image datasets.                       |
| [13] | 2019 | 15 Real-Life Datasets                                                                  | Oversampling and Undersampling Techniques               | Analyzed performance of sampling techniques across various classifiers.                                           | Did not delve into feature-level augmentation or generative models.              |
| [14] | 2020 | Various Image datasets                                                                 | KA-Ensemble (Kernel -ADASYN with Ensemble Learning)     | Combined over- and under-sampling with ensemble methods for improved classification.                              | Complexity in implementation; limited to specific scenarios.                     |
| [15] | 2019 | Breast Cancer Diagnosis Dataset                                                        | Clustering-based Undersampling with Random Oversampling | Improved classification accuracy using hybrid sampling methods.                                                   | Specific to medical datasets; generalizability is uncertain.                     |
| [16] | 2023 | Pima Indian Diabetes, Breast Cancer Wisconsin, and Insurance Claim Prediction Datasets | SMOTE, Borderline-SMOTE, ADASYN                         | Compared various oversampling techniques across classifiers.                                                      | Did not incorporate generative models or feature-level augmentations.            |
| [17] | 2023 | Heart Disease Dataset                                                                  | SMOTE, ADASYN                                           | Evaluated effectiveness of oversampling methods in medical data.                                                  | Focused solely on tabular data; image data not considered.                       |
| [18] | 2025 | Semiconductor Manufacturing Data                                                       | GANs, VAEs, Diffusion Models                            | Demonstrated robustness of generative models in virtual metrology.                                                | Domain-specific; applicability to general image classification is limited.       |
| [19] | 2020 | Medical Imaging Data                                                                   | Diffusion Models                                        | Showed potential of diffusion models in augmenting medical images.                                                | Early study; lacked extensive evaluations.                                       |
| [20] | 2024 | Medical Image Datasets                                                                 | Disc-VAE, GAN                                           | Compared VAE and GAN for medical image generation.                                                                | Limited to medical imaging; generalizability is uncertain.                       |
| [21] | 2024 | Medical Imaging Data                                                                   | Diffusion Models                                        | Highlighted the role of diffusion models in modern image augmentation.                                            | Focused on medical imaging; other domains not considered.                        |
| [22] | 2022 | Biomedical Data                                                                        | Crossover-based Data Augmentation                       | Proposed a novel data augmentation technique for biomedical data.                                                 | Specific to biomedical domain; applicability to general image data is uncertain. |
| [23] | 2022 | Recommendation System Data                                                             | Hybrid GAN-based Approach                               | Addressed imbalanced data in recommendation systems using GANs.                                                   | Domain-specific; generalization to image data is limited.                        |
| [24] | 2022 | Intrusion Detection Data                                                               | KNN and GAN-based Hybrid Approach                       | Improved classification in imbalanced intrusion detection datasets.                                               | Focused on intrusion detection; image data not considered.                       |

methods. Raj et al. [22] proposed a crossover-based data augmentation technique applied to biomedical datasets, achieving improved classification accuracy. Supporting hybridization, Shafqat and Byun [23] introduced a GAN-based hybrid approach addressing class imbalance in recommendation systems, significantly boosting prediction accuracy. However, they highlighted domain-specific limitations and suggested expanding their methods to general visual data contexts. In another hybrid approach, Ding et al. [24] combined KNN with GANs to enhance intrusion detection datasets' performance. They achieved notable accuracy improvements but acknowledged domain specificity as a significant drawback. Similarly, Strelcenia and Prakoonwit [28] surveyed GAN-based techniques for credit card fraud detection, affirming the efficacy of hybrid generative approaches yet noting a clear absence of methods integrating feature-level oversampling. Temraz and Keane [29] offered an alternative counterfactual-based augmentation method showing promise

for addressing imbalance but again emphasized the need for empirical validation across varied datasets. Dina et al. [30] demonstrated synthetic data's effectiveness for intrusion detection classifiers but recognized limited exploration in visual datasets. Nie et al. [31] specifically tackled imbalanced sky image datasets for photovoltaic prediction using CNNs with resampling and augmentation methods. They highlighted improvements yet recommended further exploration with generative models for broader applicability. Lastly, Chao and Zhang [32] introduced data augmentation for few-shot classification in imbalanced datasets, showing potential but still calling for comprehensive hybrid methods combining pixel-level and feature-level augmentation.

#### D. APPLICATIONS OF AUGMENTATION TECHNIQUES

Practical applications of balanced datasets reinforce the critical importance of addressing class imbalance effectively. Allal et al. [33] utilized balanced datasets for smart

grid stability evaluation, significantly improving predictive accuracy. Kaur et al. [34] provided a systematic review identifying class imbalance challenges across numerous machine learning applications, emphasizing the necessity for advanced, hybrid augmentation methods. Thabtah et al. [35] empirically evaluated class imbalance across various classifiers and datasets, confirming traditional methods' limitations and advocating for integrative approaches. Extending these practical insights, Sambasivam and Opiyo [36] demonstrated CNN-based cassava disease detection using balanced agricultural datasets, significantly enhancing diagnostic accuracy. Mienye and Sun [37] compared cost-sensitive methods for imbalanced medical data, highlighting critical performance improvements but suggesting hybrid augmentation for further gains. Lastly, Zhang et al. [38] introduced novel evaluation metrics for side-channel analysis, illustrating substantial improvement in imbalanced classification yet identifying remaining gaps effectively addressed by hybrid pixel-feature augmentation strategies.

### E. RESEARCH GAP AND FINDINGS

Collectively, the studies reviewed underline several critical gaps in current augmentation strategies for addressing class imbalance. Primarily, classical oversampling methods such as SMOTE and ADASYN tend to generate synthetic samples that lack genuine structural variability, often leading to limited feature-level diversity and potential overfitting. Conversely, generative approaches, including GANs, VAEs, and diffusion models, produce realistic synthetic images but frequently fall short in explicitly enhancing feature-level diversity critical for effective classifier training. Furthermore, hybrid approaches reviewed generally demonstrate promising results but remain highly domain-specific and lack broad applicability to diverse visual classification scenarios. To address these shortcomings, our proposed hybrid augmentation approach, integrating Stable Diffusion and SMOTE, effectively bridges the identified gaps. By combining Stable Diffusion's ability to generate high-quality, structurally diverse synthetic images with SMOTE's capability for creating diverse feature-level representations, our method ensures comprehensive augmentation both at the pixel and feature levels. Consequently, this integrated framework not only enhances classifier robustness but also provides broader applicability across various domains, overcoming domain-specific limitations encountered in previous hybrid methodologies.

### III. METHODS AND MATERIAL

A comprehensive methodology is developed to address severe class imbalance in visual classification tasks. The proposed pipeline shown in Fig. 2 integrates classical and modern augmentation techniques to enhance both pixel-level and feature-level diversity. Beginning with systematic data collection and preprocessing, the approach leverages advanced generative modeling to synthesize minority class images, followed by robust feature extraction using a deep convolutional



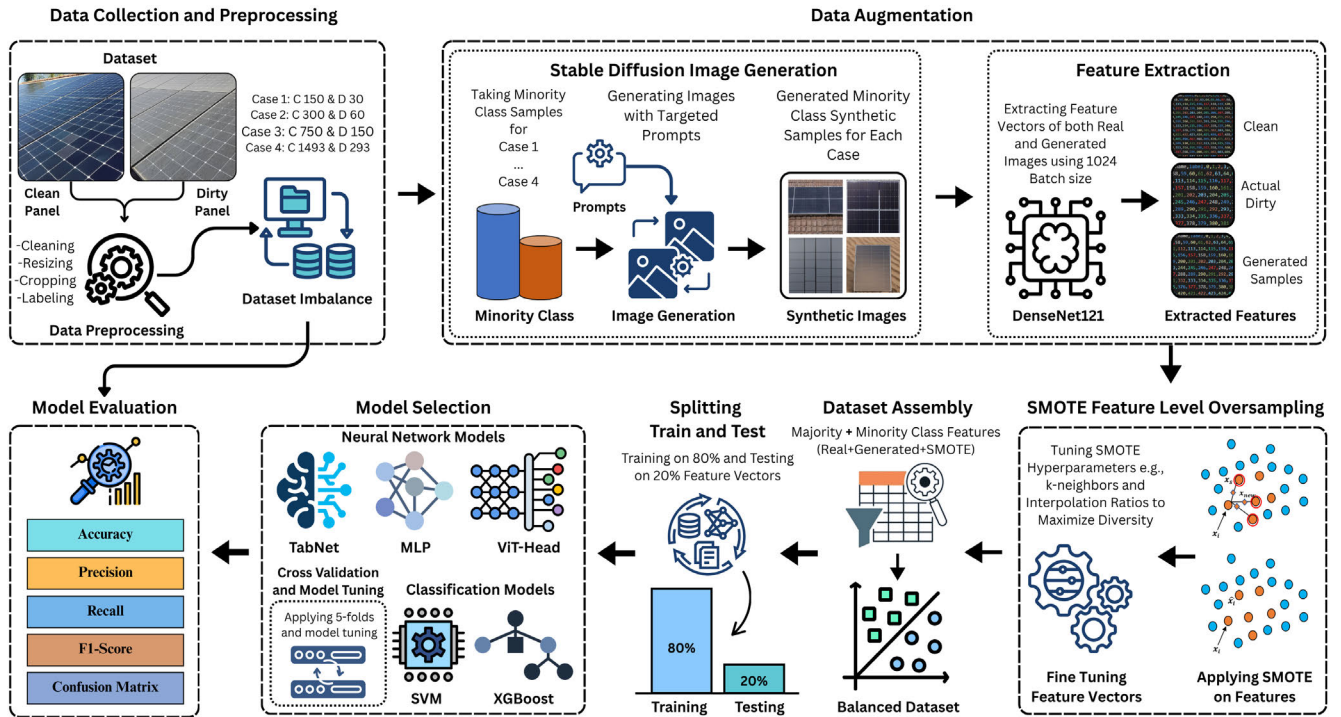
**FIGURE 1.** Solar panel dust detection dataset with clean and dirty panels taken from Kaggle [39].

backbone. To further expand the representation of minority classes, synthetic feature vectors are generated through feature-level oversampling. Finally, the enriched dataset is used to train and evaluate multiple machine learning models, ensuring a fair and thorough assessment of performance across all classes. Fig. 2 presents an overview of our proposed hybrid data augmentation pipeline, which systematically addresses class imbalance in visual classification. The workflow begins with data collection and preprocessing, followed by targeted image generation for minority classes using Stable Diffusion, and deep feature extraction via DenseNet121. To further enhance feature-level diversity, SMOTE is applied, creating additional synthetic minority class samples in feature space. The resulting features are then assembled into a balanced training dataset, enabling robust model training and evaluation.

### A. DATA ACQUISITION

The initial phase of the pipeline involves systematic data collection and preprocessing, which are critical for building a high-quality training set and ensuring the reliability of downstream modeling. For this study, we utilized a publicly available solar panel image dataset (shown in Fig. 1) from Kaggle [39], comprising a total of 2,562 images: 1,493 clean panels and 1,069 dirty panels.

While the original dataset is relatively balanced, for a thorough evaluation of our proposed hybrid augmentation approach, Diffusion-SMOTE Hybrid Augmentation (DiSMHA), it was essential to simulate a variety of class imbalance scenarios. Therefore, we considered an experimental setting with a total sample pool of 3,000 images, comprising 1,500 clean and 1,500 dirty panel samples, and systematically constructed four distinct imbalance cases. The rationale for designing these four cases was to rigorously examine DiSMHA's effectiveness across a spectrum of imbalance levels, closely mimicking the class distribution challenges encountered in real-world fault detection



**FIGURE 2.** Proposed DiSMHA Pipeline: Solar panel images are preprocessed and split into multiple imbalance cases. Minority class samples are augmented using Stable Diffusion, features are extracted via DenseNet121, and SMOTE is applied at the feature level. The combined dataset is used to train and evaluate classification models (SVM, XGBoost, MLP, ViT-Head, TabNet) with standard performance metrics.

scenarios that are shown in Fig. 3. By doing so, we are able to assess the robustness and adaptability of the augmentation pipeline under progressively more challenging minority class conditions.

#### Case 1: Severe Imbalance (Low Sample Size)

- Total Samples: 300 (10% of total selected dataset)
- Clean: 150 and Dirty 30 (10% of total samples)

#### Case 2: Moderate Imbalance (Expanded Sample Size)

- Total Samples: 600 (20% of total selected dataset)
- Clean: 300 and Dirty 60 (10% of total samples)

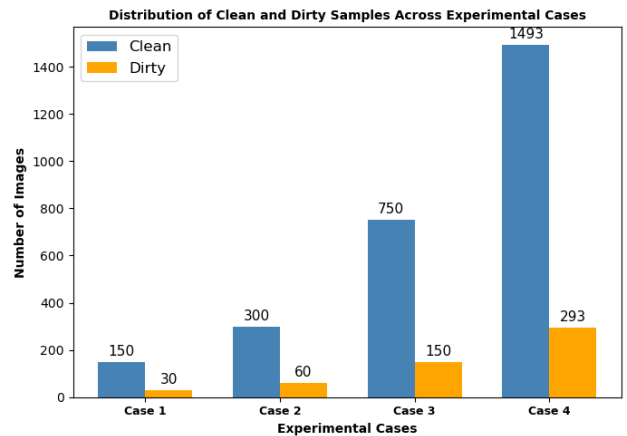
#### Case 3: Imbalance at Larger Scale

- Total Samples: 1500 (50% of total selected dataset)
- Clean: 750 and Dirty 150 (10% of total samples)

#### Case 4: Higher Minority Proportion

- Total Samples: 2986
- Clean: 1493 and Dirty 293 (10% of total samples)

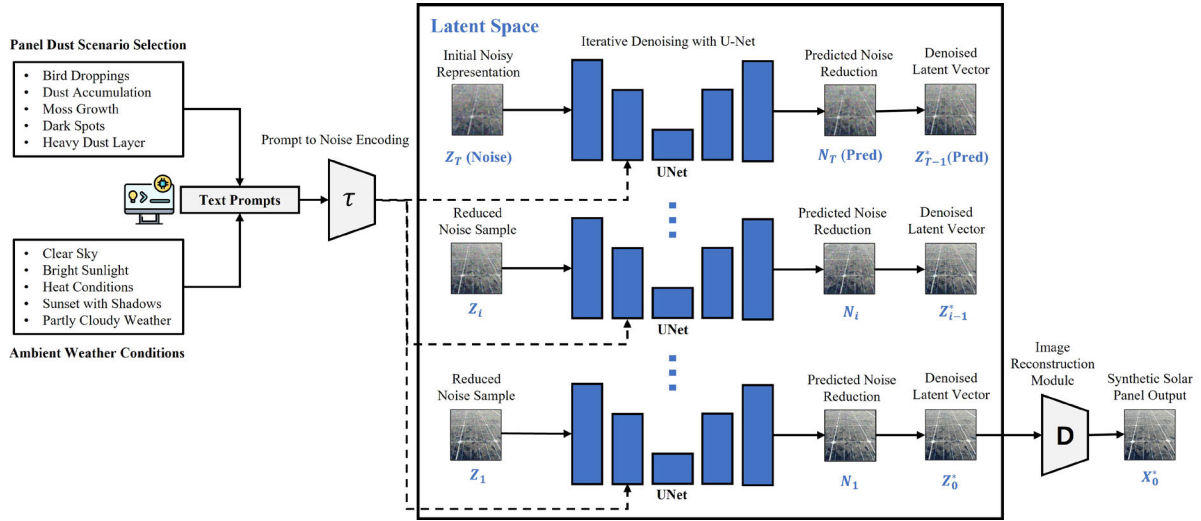
These cases were deliberately created to check both severe and moderate imbalance ratios, as well as different dataset scales, reflecting a range of operational scenarios. By systematically evaluating DiSMHA across these cases, we are able to provide a comprehensive analysis of its augmentation effectiveness based on minority class recall and generalization capability under varied real-world conditions. This multi-case approach not only highlights the strengths of our hybrid augmentation strategy but also establishes a robust benchmark for future comparative studies on data imbalance handling in solar panel defect and dust detection.



**FIGURE 3.** Distribution of clean and dirty solar panel images across the four experimental cases.

## B. DATA PRE-PROCESSING TECHNIQUES

To ensure data consistency and optimize model performance, a sequence of preprocessing steps was precisely applied to the solar panel dust detection dataset. All solar panel images were first reviewed and cleaned to remove corrupted or irrelevant samples, ensuring data integrity across the dataset. Each image was then resized to a fixed resolution of  $224 \times 224$  pixels to maintain consistency with standard deep convolutional network input requirements. Images were labeled according to their respective classes ("clean" or



**FIGURE 4.** Stable Diffusion-based pipeline for generating realistic minority class images. Text prompts are encoded, iteratively denoised in latent space, and decoded to produce high-quality synthetic samples for data augmentation.



**FIGURE 5.** Dirty panels generated through stable diffusion using specific weather conditions and dust prompts.

“dirty”) based on the original annotations. For spatial clarity, images were cropped to emphasize the solar panel area, eliminating extraneous backgrounds or borders that might otherwise introduce confusion during model training. All images were then converted to RGB format and normalized to the  $[0, 1]$  range to stabilize model learning and ensure compatibility with downstream feature extraction, following best practices in deep learning pipelines. These combined preprocessing strategies produce high-quality, standardized, and semantically relevant inputs, forming a robust foundation for both real and synthetic datasets in the detection of dust accumulation on solar panels [40].

### C. STABLE DIFFUSION BASED IMAGE GENERATION

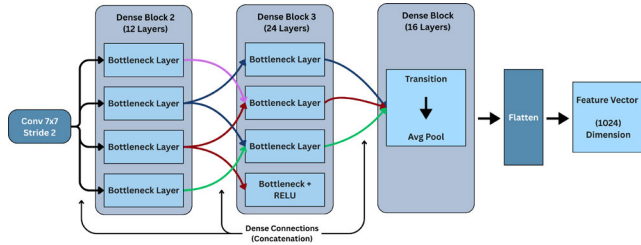
Stable Diffusion-based image generation leverages advanced generative modeling to create realistic, class-consistent synthetic images for minority classes [41]. By translating carefully engineered text prompts into diverse solar panel scenarios, this approach significantly enriches pixel-level diversity and directly addresses class imbalance within the training dataset, enhancing model robustness [41].

In this research, we designed targeted prompts representing various dirty panels and weather conditions to guide the diffusion model in producing high-quality, minority-class images. The generated outputs were then filtered through

automated and manual quality control to ensure semantic consistency and visual realism. These synthetic images were subsequently integrated into the training pipeline to augment minority class representation. Our hybrid augmentation framework (DiSMHA) leverages a Stable Diffusion-based pipeline to synthesize high-quality, class-consistent images for minority (dirty panel) classes, as illustrated in Fig. 4 and integrated within the overall workflow in Fig. 2. The process begins with prompt engineering, wherein semantic text prompts are carefully constructed by combining domain-specific panel dust conditions (e.g., bird droppings, dust accumulation) and various weather scenarios. These prompts, shown in the left section of Fig. 4, are mapped into the latent space via the encoder module, injecting controlled randomness and condition-specific semantics. Within the latent space, the model iteratively refines a noisy latent representation  $Z_T$  through a sequence of denoising U-Net blocks. At each stage, predicted noise is subtracted, and denoised intermediates  $Z_{T-1}^*$ ,  $Z_{i-1}^*$ ,  $Z_0^*$  are generated, as depicted in the central block of Fig. 4. This process enables the model to progressively recover structured, visually meaningful panel images aligned with the initial prompts. The diffusion process works by adding noise to the input image in several steps and reversing this process using a U-Net decoder shown in Fig. 4. For each step, the model predicts the noise in the image and refines it by using Equation (1) for this process is as follows:

$$Z_T = Z_{T-1} + Z_{(T-1)\text{NoisePredictions}} \quad (1)$$

where  $Z_T$  is the noise image at step  $T$ , and  $Z_{T-1}\text{NoisePredictions}$  is the predicted noise used to refine the image. This process continues until the image reaches the final step, where the noise is fully removed, and the image is returned to its final latent space representation  $Z_0^*$  as shown in Fig. 4. Image synthesis



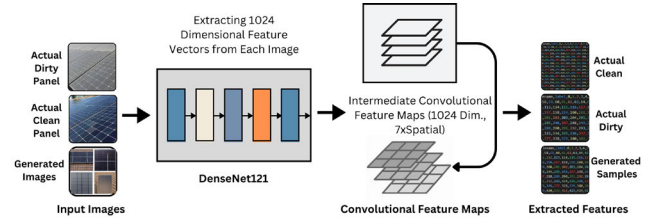
**FIGURE 6.** DenseNet121 architecture utilizing the densely connected convolutional blocks and bottleneck layers, followed by global average pooling and flattening, to generate compact 1024-dimensional feature vectors.

for minority classes is achieved by targeting underrepresented conditions during prompt construction, ensuring diversity and relevance among generated samples [42]. Finally, the synthetic images are generated by following the mentioned pipeline based on prompts and the given weather conditions shown in Fig. 5. This pipeline significantly enriches the minority class representation, maintaining data diversity and supporting robust classifier performance across imbalance scenarios.

#### D. FEATURE EXTRACTION

For robust downstream classification, we employed DenseNet121 as our convolutional neural network backbone to extract informative and discriminative features [43] from all input images, including real clean panels, real dirty panels, and synthetic samples generated via Stable Diffusion as shown in Fig. 2. As illustrated in Fig. 6, each input image, regardless of origin, is processed through the DenseNet121 model, which outputs a 1024-dimensional feature vector. This high-level embedding encapsulates salient spatial and structural information critical for distinguishing between classes.

The internal architecture of DenseNet121, shown in Fig. 6, comprises a series of densely connected convolutional blocks. Each layer receives feature maps from all preceding layers, facilitating enhanced feature reuse and gradient flow. Following a series of bottleneck and transition layers, global average pooling is applied to the final convolutional feature maps, resulting in a fixed-length 1024-dimensional vector for each image [44]. These vectors serve as the extracted feature representations that are used in subsequent steps, such as SMOTE-based feature-level oversampling and model training [45]. This feature extraction procedure ensures that both real and synthetic images are transformed into a unified, high-dimensional feature space, enabling consistent downstream processing and fair evaluation of augmentation effects. The extracted features, as visualized in Fig. 7, provide the foundation for class balancing and classification in our hybrid pipeline illustrated in Fig. 2. DenseNet121 was chosen as the feature extraction backbone due to its proven efficiency in capturing deep hierarchical features with reduced redundancy through dense connectivity, outperforming many traditional CNNs in both accuracy and computational cost for mid-sized image datasets. For SMOTE, the number of



**FIGURE 7.** Feature extraction pipeline using DenseNet121 to obtain 1024-dimensional vectors from real and generated solar panel images.

nearest neighbors ( $k$ ) was set to 5, based on grid search for optimal balance between synthetic diversity and overfitting risk. The sampling strategy was “auto” to ensure minority classes were matched to the majority. For Stable Diffusion, key parameters such as diffusion steps (set to 50) and noise levels (standard deviation 0.7) were selected via empirical validation, prioritizing a trade-off between image realism and diversity. These choices were made to maximize both feature and pixel-level diversity while maintaining computational efficiency and model generalization.

#### E. SMOTE FEATURE-LEVEL OVERSAMPLING

After extracting high-dimensional feature vectors from both real and generated images using DenseNet121, all features are merged into a unified feature set for each experimental case Fig. 3. This consolidated feature space includes actual clean panels, actual dirty panels, and synthetic samples, providing a rich input for class balancing. The Synthetic Minority Over-sampling Technique (SMOTE) is then applied at the feature level, as visually illustrated in Fig. 8 and further detailed in Fig. 9 and the main workflow (Fig. 2). Each minority class feature vector (dirty panel) is treated as a point in high-dimensional space. For each, SMOTE randomly selects a sample  $x_i$  and identifies its  $k$  – nearest neighbors among the minority class (dirty) samples. New synthetic samples are generated by interpolating between  $x_i$  and its neighbor  $\hat{x}_i$  as shown in below Equation (2):

$$x_{\text{new}} = x_i + \delta (\hat{x}_i - x_i), \quad \delta \in [0, 1] \quad (2)$$

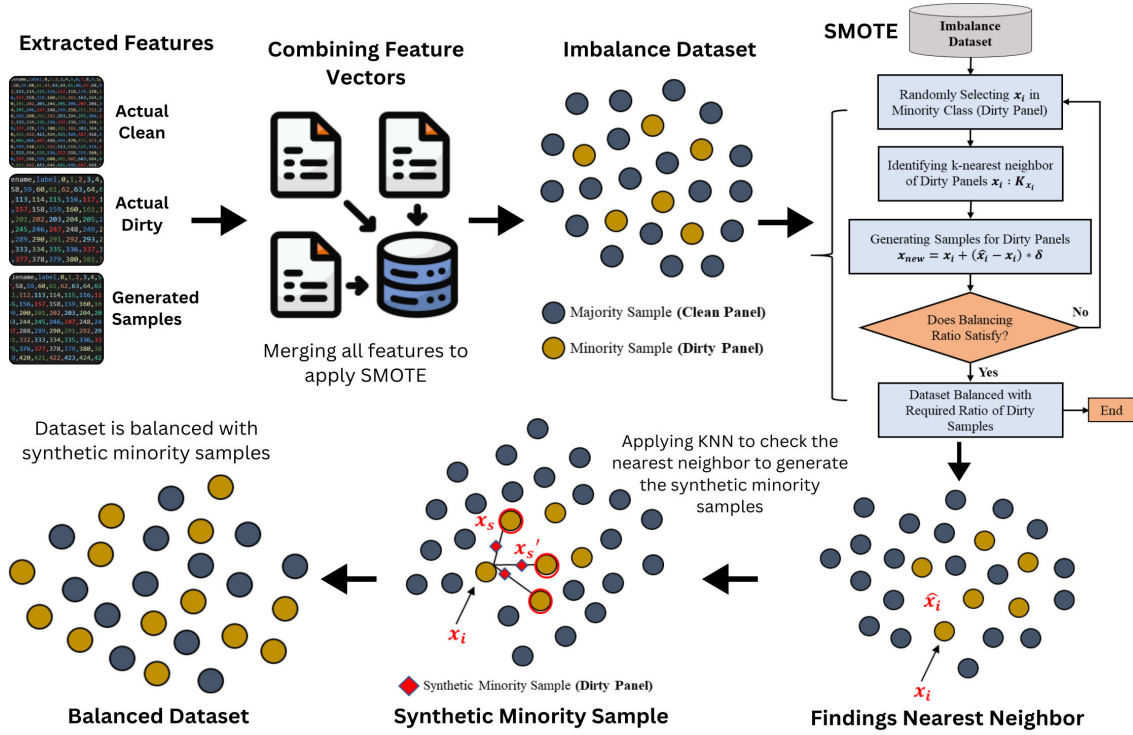
where:  $x_{\text{new}}$  is the synthetic sample,

$\hat{x}_i$  is a randomly selected neighbor of  $x_i$ .

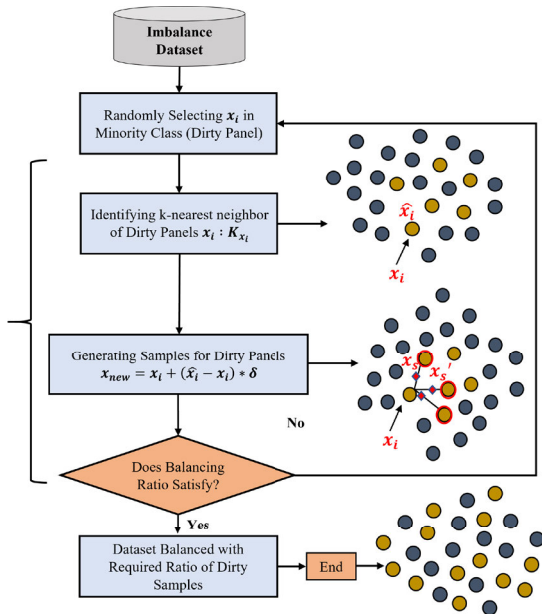
$\delta$  is a random number between 0 and 1 that controls the interpolation.

This procedure is repeated until the desired balancing ratio is achieved, as depicted in the flowcharts and scatter plots within the attached figures.

The generated synthetic feature vectors are explicitly labeled and integrated with the original dataset to correct the class imbalance. SMOTE is performed independently for each of the four experimental cases discussed in Fig. 3. In each case, the procedure ensures that the final training set has equal representation for clean and dirty panels, thus enabling fair and robust model training. This SMOTE-driven feature-level oversampling (see Fig. 9 and 8) is a crucial step in our hybrid augmentation pipeline, ensuring



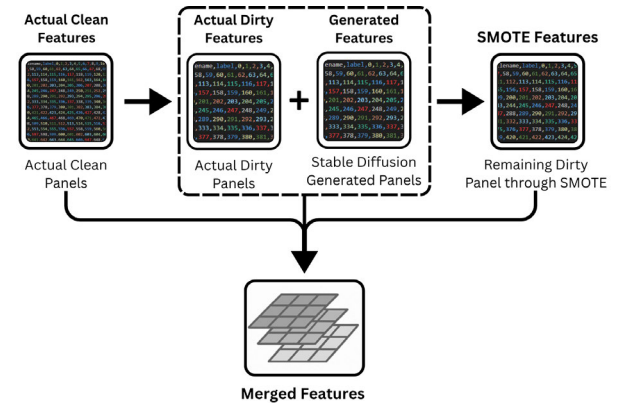
**FIGURE 8.** SMOTE Feature-Level Oversampling Workflow: Extracted feature vectors from clean, dirty, and generated images are merged and analyzed in feature space. SMOTE generates synthetic minority samples by interpolating between neighbors, balancing the dataset for robust classification.



**FIGURE 9.** SMOTE algorithm for feature-level oversampling. For each minority (dirty panel) sample, k-nearest neighbors are identified, synthetic samples are generated by interpolation, and the process is repeated until the dataset is balanced.

effective mitigation of class imbalance and enhancing model generalizability across varying data regimes.

All these feature sets are merged, after which SMOTE-generated synthetic minority (dirty panel) features are incorporated to address any remaining imbalance. The



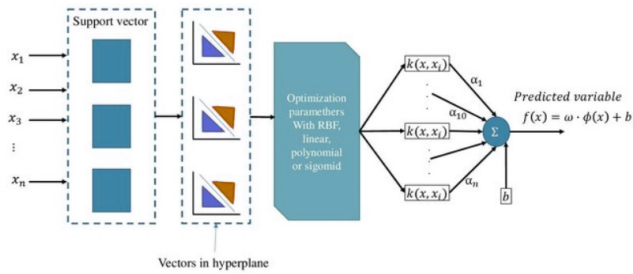
**FIGURE 10.** Assembling the final balanced feature by merging actual clean, actual dirty, Stable Diffusion-generated, and SMOTE-augmented features for model training.

assembly process in Fig. 10 ensures that the resulting dataset achieves a balanced representation of both classes.

## F. MODEL SELECTION

### 1) SVM

Support Vector Machines (SVMs) are powerful supervised learning algorithms designed for robust classification, particularly in high-dimensional spaces [46]. As shown in Fig. 11, SVMs operate by projecting input feature vectors onto a hyperplane and optimizing parameters using kernel functions such as RBF, linear, polynomial, or sigmoid [47]. The decision function aggregates these optimized vectors



**FIGURE 11.** Schematic of the Support Vector Machine (SVM) classification process, illustrating projection of feature vectors into a hyperplane and prediction using optimized kernel functions.

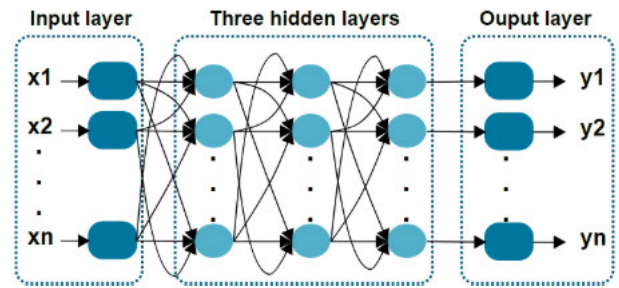
to predict class membership, enabling efficient discrimination even in non-linear scenarios. In our research, SVM was chosen due to its proven effectiveness in handling imbalanced and complex datasets, especially after feature augmentation [48]. In a recent study, Wilson et al. [49] demonstrate that SVMs excel in scenarios involving synthetic oversampling and high-dimensional feature spaces, offering strong generalization and resilience to noise. By leveraging kernel-based optimization, SVM efficiently separates classes within our enriched, balanced feature set, supporting reliable solar panel condition classification [50].

## 2) MLP

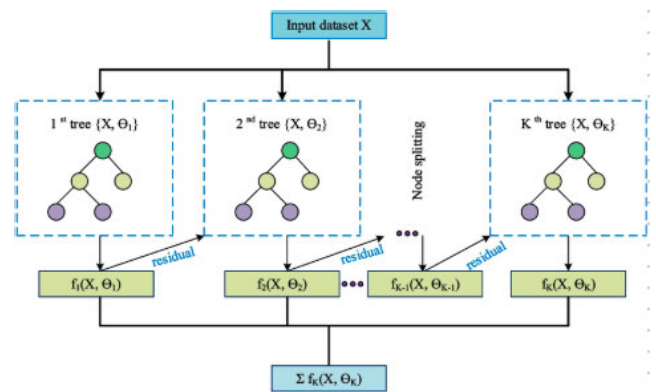
Multilayer Perceptrons (MLPs) are fundamental feedforward artificial neural networks consisting of an input layer, one or more hidden layers, and an output layer [51], as depicted in Fig. 12. Each neuron in a layer is fully connected to every neuron in the subsequent layer, allowing for the modeling of complex, non-linear relationships in data. The network learns optimal weights through backpropagation, enabling the transformation of input features ( $x_1, x_2, \dots, x_n$ ) into predictive outputs ( $y_1, y_2, \dots, y_n$ ) via successive hidden layers. MLPs are particularly suited for high-dimensional feature spaces, such as those derived from deep feature extraction and synthetic augmentation [52]. In our research, we employ MLPs due to its ability to integrate and learn from the diverse, enriched feature sets generated by both Stable Diffusion and SMOTE. In a recent study, Sharma et al. [53] confirm the effectiveness of MLPs in image-based and imbalanced classification problems, especially when combined with advanced data augmentation techniques. Thus, the MLP serves as a robust baseline for evaluating the discriminative power of our augmented dataset.

## 3) XGBoost

Extreme Gradient Boosting (XGBoost) is a highly efficient ensemble learning algorithm based on gradient-boosted decision trees [54]. As illustrated in Fig. 13, XGBoost sequentially constructs multiple decision trees, where each subsequent tree learns from the residuals (errors) of its predecessors. The outputs of all trees are aggregated to form the final prediction, enabling the model to capture complex, non-linear patterns in the data. Node splitting and



**FIGURE 12.** Architecture of a Multilayer Perceptron (MLP) with input, multiple hidden, and output layers, illustrating fully connected neuron pathways for feature transformation and prediction.



**FIGURE 13.** Architecture of XGBoost: sequential construction of decision trees, each learning from the residuals of previous trees, with outputs aggregated for final prediction.

optimization occur iteratively, with each tree contributing incrementally to improved accuracy. XGBoost is renowned for its robustness, scalability, and superior performance on structured datasets, particularly in the presence of class imbalance [55].

In our research, XGBoost is employed to leverage its capacity to handle high-dimensional and synthetically balanced feature spaces derived from DenseNet121, Stable Diffusion, and SMOTE. Imani et al. [56] have demonstrated the effectiveness of XGBoost in imbalanced and augmented data scenarios, where it outperforms many traditional classifiers in terms of precision, recall, and overall robustness. This makes XGBoost a suitable choice for our solar panel defect classification task.

## 4) ViT-HEAD

Vision Transformer (ViT) heads represent a breakthrough in computer vision by adapting transformer architectures—originally designed for natural language processing—to image classification tasks [57]. Unlike traditional convolutional neural networks, ViT splits input images into fixed-size patches, linearly embeds each patch, and processes the sequence using self-attention mechanisms [58]. This enables the model to capture both local and global dependencies within the data, leading to enhanced feature representation and discrimination, especially in complex or diverse

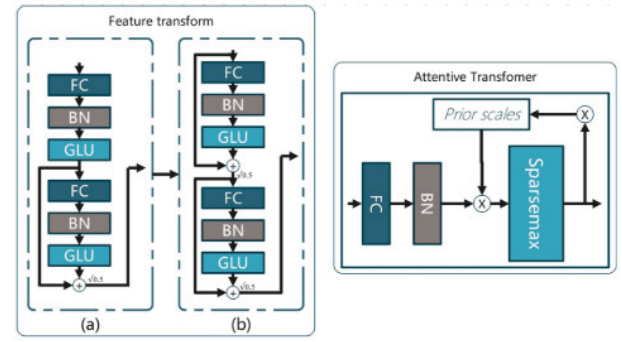
image datasets. In our research, the ViT-Head is integrated to exploit its superior ability to model intricate spatial relationships present in high-dimensional features derived from real, synthetic, and SMOTE-augmented data. ViT architectures have shown state-of-the-art results in a variety of image classification benchmarks and are particularly effective when training data is enriched via data augmentation strategies [59]. The model's inherent scalability and robustness to class imbalance make it especially suitable for our solar panel dust classification task, where capturing subtle and distributed cues is critical for distinguishing between clean and dirty panels [60]. By leveraging ViT-Head alongside other classifiers, we provide a comprehensive evaluation of our hybrid augmentation pipeline's effectiveness.

### 5) TabNet

TabNet is a cutting-edge deep learning architecture specifically designed for learning from tabular data, combining the strengths of sequential feature selection and attention mechanisms [61]. As illustrated in Fig. 14, TabNet's architecture consists of two main components: feature transformation blocks and attentive transformers. The feature transformer (left panel) applies sequential fully connected (FC), batch normalization (BN), and gated linear unit (GLU) layers, with residual connections to stabilize learning and enable effective information flow. The attentive transformer (right panel) utilizes attention scores, computed via sparsemax and modulated by prior scales, to selectively focus on the most relevant features at each decision step. This unique design allows TabNet to perform dynamic feature selection while maintaining model interpretability and robust performance [62]. In our research, TabNet is integrated as a classifier to leverage its ability to handle high-dimensional and heterogeneously augmented feature spaces, which include real, generated, and SMOTE-augmented features. Its inherent interpretability enables us to identify which features, either from real images or synthetic augmentations are most critical for accurate solar panel dust classification [63]. Awuku et al. [64] have demonstrated TabNet's strong performance on imbalanced, augmented, and multi-source tabular data, making it a highly suitable choice for our hybrid augmentation pipeline. This facilitates fair comparison and robust classification in scenarios where traditional neural networks or tree-based models may struggle with feature redundancy or class imbalance.

### G. PERFORMANCE EVALUATION METRICS

To comprehensively assess the performance of our classifiers on the balanced and augmented datasets, we employ several widely accepted evaluation metrics: accuracy, precision, recall, and F1-score. These metrics provide a robust understanding of not only the overall correctness of the models but also their ability to handle class imbalance and distinguish between minority and majority classes [65].



**FIGURE 14.** TabNet architecture: (a) and (b) show sequential feature transformation blocks with fully connected (FC), batch normalization (BN), and gated linear unit (GLU) layers with residual connections with an attentive transformer for dynamic feature selection and prior scale modulation.

### 1) ACCURACY

Accuracy measures the proportion of correctly predicted samples out of the total predictions, serving as an overall indicator of model performance [66]. While accuracy is intuitive and provides a high-level view, it can be misleading for imbalanced datasets since high accuracy may be achieved by favoring the majority class. In this research, accuracy is used alongside other metrics to provide context for overall classifier correctness, but is not the sole criterion for performance evaluation. It is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

### 2) PRECISION

Precision quantifies the proportion of true positive predictions among all samples predicted as positive (e.g., dirty panels) [66]. This metric is essential in applications where false positives are costly, such as incorrectly identifying a clean panel as dirty, which could lead to unnecessary maintenance. By focusing on the reliability of positive predictions, precision helps us evaluate the classifier's trustworthiness in real-world scenarios, making it a suitable metric for our defect detection pipeline. It is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

### 3) RECALL

Recall measures the proportion of actual positive samples (e.g., truly dirty panels) that are correctly identified by the model [67]. High recall indicates that the classifier is effective at capturing most of the minority class, minimizing false negatives. In our study, recall is especially important for ensuring that dust-affected panels are not missed, thus supporting timely maintenance and maximizing solar efficiency. It is given by:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

**TABLE 2. Computational environment and resources used in proposed research.**

| Computing Resources | Detail                                       |
|---------------------|----------------------------------------------|
| CPU                 | Intel(R) Core(TM) i7-13700 (24 CPUs) 2.1 GHz |
| GPU                 | NVIDIA GeForce RTX 4060                      |
| RAM                 | 16 GB                                        |
| System              | 64-bit operating system, x64-based processor |
| Language            | Python(3.9.19)                               |
| Framework           | Keras(2.7.0) with TensorFlow(2.7.0) backend  |
| Software            | VisualStudio                                 |

#### 4) F1 SCORE

The F1-score is the harmonic mean of precision and recall, providing a balanced assessment that accounts for both false positives and false negatives [68]. This metric is particularly useful in the presence of class imbalance, as it avoids inflated scores from one-sided performance. In this research, the F1-score serves as a key criterion for comparing models and augmentation strategies, ensuring robust detection of minority class instances without compromising precision or recall. The F1-Score is the harmonic mean of precision and recall and is defined as:

$$\text{F1 Score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}} \quad (6)$$

## IV. RESULTS

This section presents a comprehensive evaluation of our proposed hybrid augmentation framework, DiSMHA, across all experimental cases and classifier models. We report the performance metrics for each approach, highlighting the effectiveness of stable diffusion, SMOTE, and their integration in addressing severe class imbalance. Comparative analyses are provided to demonstrate improvements in accuracy, precision, recall, F1-score, and confusion matrix outcomes for both the majority and minority classes in Section V. The results underscore the benefits of our pipeline in enhancing the detection of dust-affected solar panels and validate the robustness of each classification model on augmented datasets.

### A. EXPERIMENTAL SETUP

#### 1) COMPUTATIONAL ENVIRONMENT AND RESOURCES

In this research, all experiments were conducted on a high-performance computing setup equipped with an Intel(R) Core(TM) i7-13700 CPU (24 cores, 2.1 GHz), 16 GB RAM, and an NVIDIA GeForce RTX 4060 GPU, as detailed in Table 2. The system runs a 64-bit operating system with Python (3.9.19) as the primary language. Deep learning models were implemented using Keras (2.7.0) with a TensorFlow (2.7.0) backend. The software environment also includes Visual Studio for code development and popular Python libraries such as NumPy, Pandas, Matplotlib, and scikit-learn. SMOTE from the imblearn library was employed for feature-level oversampling and dataset balancing.

#### 2) DATA SPLIT FOR TRAINING AND TESTING

For all experimental cases, the dataset was divided into distinct training and testing sets to ensure unbiased evaluation.

Specifically, 80% of the data was allocated for training, with the remaining 20% reserved for testing. This split was consistently applied across all imbalance cases and augmentation scenarios, ensuring comparability of results. During model training, early stopping was employed with a patience criterion, allowing training to proceed for up to 100 epochs but halting automatically if validation performance plateaued, thereby reducing overfitting and improving accuracy. Cross-validation was also utilized during model tuning to further enhance the robustness and reliability of performance assessments.

#### 3) HYPERPARAMETER TUNING

For each classifier, hyperparameter tuning was explicitly performed using grid search to identify optimal parameter settings. This approach systematically evaluated predefined parameter grids for models including SVM (kernel type, C, gamma), MLP (layer size, learning rate), XGBoost (tree depth, learning rate), ViT-Head (attention heads, embedding size), and TabNet  $n_{steps}$  and  $n_d$ .

### B. PERFORMANCE EVALUATION WITHOUT AUGMENTATION

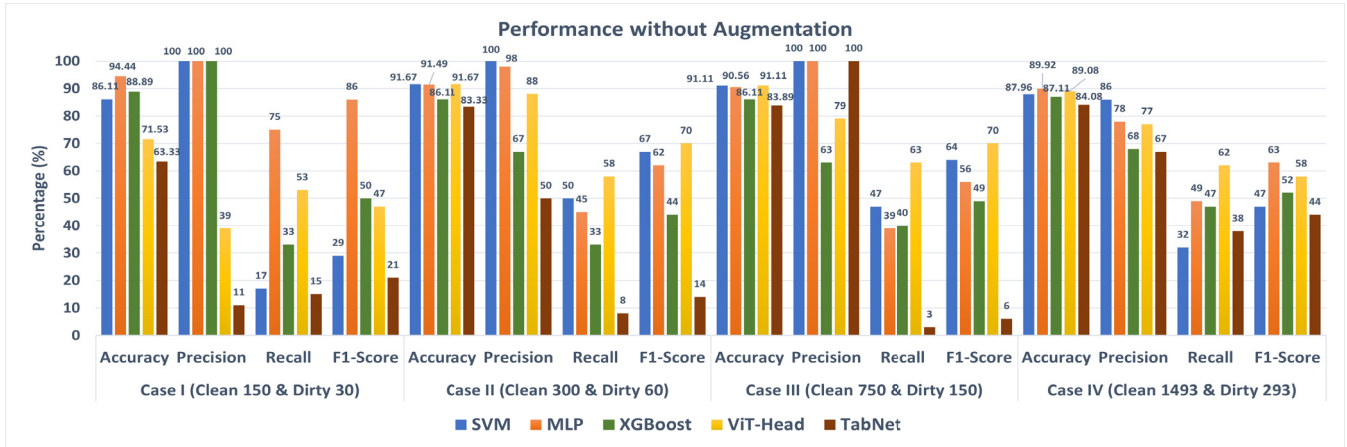
The results in Fig. 15 present the classification performance of SVM, MLP, XGBoost, ViT-Head, and TabNet across four cases (performance results are shown in tables for each case) of increasing sample size but consistently imbalanced ratios. Each case represents a real-world scenario where the minority class (dirty panels) remains underrepresented, challenging the robustness of each classifier.

#### 1) CASE I: SEVERE IMBALANCE (CLEAN 150 AND DIRTY 30)

In case I, with only 30 dirty panels out of 180 samples, this scenario is a rigorous test for minority class detection. All the results for case I are shown in Table 3 where MLP achieves the highest accuracy (94.44%), but SVM and XGBoost also show competitive values above 86%. However, despite perfect or near-perfect precision for most models, recall is notably low SVM at just 17%, ViT-Head at 53%, and MLP the highest at 75%. This leads to modest F1 scores, where MLP stands out (86%), but most models fall below 50%. These results highlight the well-known tendency of classifiers to favor the majority class under extreme imbalance, as seen in [69], where models achieve high accuracy but fail to adequately detect minority instances. The high precision but low recall pattern is typical, indicating the models are conservative in predicting dirty panels—missing many true positives to avoid false alarms.

#### 2) CASE II: MODERATE IMBALANCE (CLEAN 300 AND DIRTY 60)

In this case, by making the dataset size double, the overall stability of data and performance is also increased, as shown in Table 3. MLP and ViT-Head both achieve strong accuracy (91.49% and 91.67%), with MLP also showing a significant boost in recall (45%) and F1-score (62). XGBoost and TabNet



**FIGURE 15.** Performance comparison without augmentation across imbalance cases, showing Accuracy, Precision, Recall, and F1-Score on solar panel dataset.

**TABLE 3.** Classification performance with all imbalanced dataset cases without augmentation.

| Case     | Model    | Accuracy | Precision | Recall | F1-Score |
|----------|----------|----------|-----------|--------|----------|
| Case I   | SVM      | 86.11    | 100       | 17     | 29       |
|          | MLP      | 94.44    | 100       | 75     | 86       |
|          | XGBoost  | 88.89    | 100       | 33     | 50       |
|          | ViT-Head | 71.53    | 39        | 53     | 47       |
|          | TabNet   | 63.33    | 11        | 15     | 21       |
| Case II  | SVM      | 91.67    | 100       | 50     | 67       |
|          | MLP      | 91.49    | 98        | 45     | 62       |
|          | XGBoost  | 86.11    | 67        | 33     | 44       |
|          | ViT-Head | 91.67    | 88        | 58     | 70       |
|          | TabNet   | 83.33    | 50        | 08     | 14       |
| Case III | SVM      | 91.11    | 100       | 47     | 64       |
|          | MLP      | 90.56    | 100       | 39     | 56       |
|          | XGBoost  | 86.11    | 63        | 40     | 49       |
|          | ViT-Head | 91.11    | 79        | 63     | 70       |
|          | TabNet   | 83.89    | 100       | 03     | 06       |
| Case IV  | SVM      | 87.96    | 86        | 32     | 47       |
|          | MLP      | 89.92    | 78        | 49     | 63       |
|          | XGBoost  | 87.11    | 68        | 47     | 52       |
|          | ViT-Head | 89.08    | 77        | 62     | 58       |
|          | TabNet   | 84.08    | 67        | 38     | 44       |

continuously struggle to improve the recall (33% and 15%, respectively), which is discussed in an already published study [70] that tree-based and attention-based models are often less sensitive to rare classes in imbalanced data without explicit balancing. Again, most models exhibit high precision but low recall, leading to suboptimal F1 scores. These trends reinforce that, although larger data helps mitigate overfitting, the intrinsic imbalance remains a primary barrier to reliable minority class recognition [53].

### 3) CASE III: INCREASED SAMPLE (CLEAN 750 AND DIRTY 150)

As the sample size increases to 900, model performances generally stabilize, as shown in Table 3. SVM and ViT-Head achieve over 91% accuracy, while MLP remains strong at 90.56%. Recall values improve for some models (SVM: 47%, MLP: 39%, ViT-Head: 63%), leading to F1-scores between 47 and 70. Notably, ViT-Head achieves the highest recall (63%) and F1-score (70), suggesting transformer-based

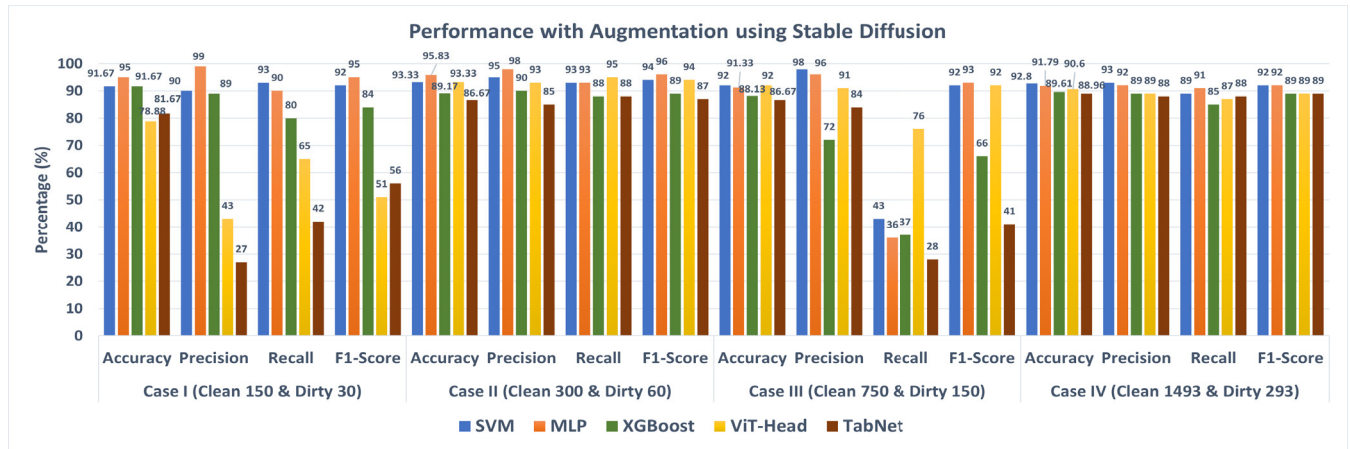
approaches may offer better generalization for minority class detection at larger scales—a trend also observed in recent literature [71]. However, TabNet’s recall and F1 remain low (6%), indicating sensitivity to minority representation and possible overfitting to the majority class.

### 4) CASE IV: LARGE IMBALANCE (CLEAN 1493 AND DIRTY 293)

In the largest dataset, all models maintain high accuracy, almost greater than 87%, but minority recall generally remains suboptimal, such as SVM (32%), MLP (49%), ViT-Head (47%) as shown in Table 3. MLP achieves the best balance with an F1-score of 63%, while TabNet again lags with 44%. This persistent recall deficit, even with more data, confirms prior findings [5] that standard classifiers alone cannot fully overcome severe class imbalance, regardless of sample size. Notably, ViT-Head and MLP show the most promise for minority class sensitivity, but overall, precision continues to outweigh recall for all methods. These results shown in Table 3 strengthen the necessity of advanced augmentation techniques such as SMOTE and stable diffusion as traditional classifiers, even with powerful architectures, consistently exhibit high accuracy and precision but limited recall for minority classes in severely imbalanced settings. This observation is well documented in recent studies [5], [53], [71] underscoring the contribution and importance of hybrid augmentation frameworks like DiSMHA in real-world defect detection scenarios.

### C. PERFORMANCE EVALUATION WITH AUGMENTATION USING STABLE DIFFUSION

The results in Fig. 16 highlight the substantial impact of Stable Diffusion-based augmentation on classification performance across all cases and models, particularly for the minority class (dirty panels). This approach consistently enhances recall and F1-score, demonstrating improved minority class detectability while maintaining high precision and accuracy [72].



**FIGURE 16.** Performance of SVM, MLP, XGBoost, ViT-Head, and TabNet across imbalanced cases, showing Accuracy, Precision, Recall, and F1-Score after augmentation with Stable Diffusion.

### 1) CASE I: SEVERE IMBALANCE (CLEAN 150 AND DIRTY 30)

In this case, with only 30 dirty panels, baseline models previously suffered from extremely low recall (SVM: 17%, ViT-Head: 53%), indicating poor minority detection as shown in Fig. 15. However, with Stable Diffusion augmentation, recall for SVM, MLP, and XGBoost jumps to 90–93%, and F1-scores reach 92–95%, as shown in Table 4. Precision remains very high (90%) across all models. These improvements show that synthetic images help overcome information scarcity, enabling classifiers to recognize rare patterns more robustly. The dramatic recall and F1-score gains address the classic challenge of imbalanced learning, consistent with findings by [73], where generative augmentation improved minority detection without sacrificing majority class precision.

### 2) CASE II: MODERATE IMBALANCE (CLEAN 300 AND DIRTY 60)

As the dataset doubles, the effect of augmentation remains impressive, as shown in Table 4. SVM, MLP, and XGBoost achieve accuracy above 93%, with recall between 93% and 88%, and F1-scores at 94–96. ViT-Head and TabNet also show strong recall (95% and 88%), a major improvement from the previous case without augmentation. These results confirm that synthetic data not only balances class distribution but also provides valuable diversity, allowing models to generalize better. This outcome aligns with [7], which reported significant recall and F1-score gains for minority classes after diffusion-based data augmentation.

### 3) CASE III: INCREASED SAMPLE (CLEAN 750 AND DIRTY 150)

In this case III, with 900 samples, augmentation continues to deliver: all models have recall values above 36%, with SVM and ViT-Head achieving 43% and 76%, respectively, and F1-scores of 92% and 96% as shown

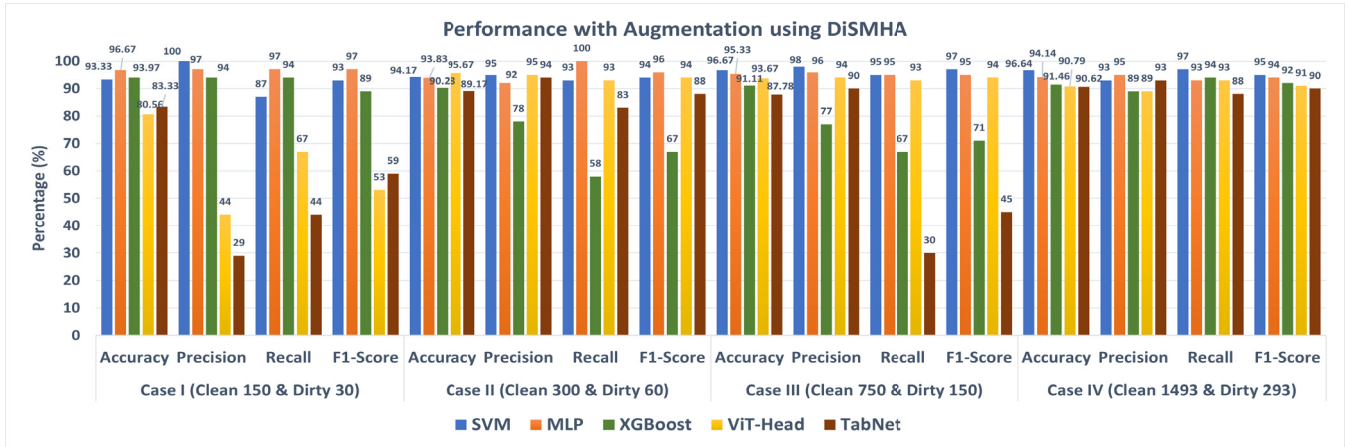
**TABLE 4.** Classification performance with all imbalanced dataset cases with augmentation using stable diffusion.

| Case     | Model    | Accuracy | Precision | Recall | F1-Score |
|----------|----------|----------|-----------|--------|----------|
| Case I   | SVM      | 91.67    | 90        | 93     | 92       |
|          | MLP      | 95.00    | 99        | 90     | 95       |
|          | XGBoost  | 91.67    | 89        | 80     | 84       |
|          | ViT-Head | 78.88    | 43        | 65     | 51       |
|          | TabNet   | 81.67    | 27        | 42     | 56       |
| Case II  | SVM      | 93.33    | 95        | 93     | 94       |
|          | MLP      | 95.83    | 98        | 93     | 96       |
|          | XGBoost  | 89.17    | 90        | 88     | 89       |
|          | ViT-Head | 93.33    | 93        | 95     | 94       |
|          | TabNet   | 86.67    | 85        | 88     | 87       |
| Case III | SVM      | 92.00    | 98        | 43     | 92       |
|          | MLP      | 91.33    | 96        | 36     | 93       |
|          | XGBoost  | 88.13    | 72        | 37     | 66       |
|          | ViT-Head | 92.00    | 91        | 76     | 92       |
|          | TabNet   | 86.67    | 84        | 28     | 41       |
| Case IV  | SVM      | 92.80    | 93        | 89     | 92       |
|          | MLP      | 91.79    | 92        | 91     | 92       |
|          | XGBoost  | 89.61    | 89        | 85     | 89       |
|          | ViT-Head | 90.60    | 89        | 87     | 89       |
|          | TabNet   | 88.96    | 88        | 88     | 89       |

in Table 4. XGBoost, MLP, and TabNet maintain high precision (72%) and F1-scores (41–92%). The results for ViT-Head are particularly notable, reflecting the advantage of combining transformer models with enhanced, diverse training data. When compared to the baseline, recall, and F1-score improvements are substantial for every model. This demonstrates that as dataset size and diversity increase, so does model robustness—consistent with trends observed by Dosovitskiy et al. [71].

### 4) CASE IV: LARGE IMBALANCE (CLEAN 1493 AND DIRTY 293)

In the largest experimental case, all models exhibit balanced and impressive performance as shown in Table 4. SVM, MLP, and ViT-Head deliver recall of 89%, with F1-scores of 92% or higher, and accuracy close to or above 91%. Even TabNet and XGBoost reach recall and F1-scores above 85%. Compared to the non-augmented scenario, where recall rarely exceeded 49% for any model, these results



**FIGURE 17.** Comparative performance of SVM, MLP, XGBoost, ViT-Head, and TabNet on all four imbalanced cases using the proposed DiSMHA hybrid augmentation, evaluated by Accuracy, Precision, Recall, and F1-Score.

validate the scalability and practical utility of diffusion-based augmentation in real-world imbalanced settings. This finding mirrors outcomes from recent research [18], underscoring that advanced augmentation not only boosts minority class recall but also preserves model precision and overall stability.

The consistently improved recall and F1-scores across all models and cases shown in Fig. 16 illustrate that Stable Diffusion augmentation addresses both data scarcity and class imbalance, enabling more equitable learning and better real-world applicability. This outcome is supported by prior studies [18], [71], [73] demonstrating that diffusion-based or generative data augmentation fills critical minority class gaps, increases sample diversity, and results in substantial performance gains—especially in imbalanced and limited-data contexts. In our research, these enhancements directly translate to more reliable detection of dust-affected solar panels, supporting proactive maintenance and energy optimization.

#### D. PERFORMANCE EVALUATION WITH HYBRID AUGMENTATION USING DISMHA

The results in Fig. 17 demonstrate the effectiveness of the proposed DiSMHA hybrid augmentation framework (combining Stable Diffusion and SMOTE) across all imbalance scenarios and classifiers. Consistently, DiSMHA leads to higher recall, F1-score, and robust overall performance—especially for models previously sensitive to minority class scarcity, such as TabNet.

##### 1) CASE I: SEVERE IMBALANCE (CLEAN 150 AND DIRTY 30)

As the results shown in Table 5 For the most imbalanced scenario, SVM achieves 93% accuracy, 87% recall, and 93% F1-score, whereas MLP achieves outstanding performance with 96.67% accuracy, 97% recall, and 97% F1-score. Even TabNet, previously limited, now delivers 44% recall and a 59% F1-score—substantially higher than its performance without augmentation (21/29%) and better

**TABLE 5.** Classification performance with all imbalanced dataset cases with augmentation using DiSMHA.

| Case     | Model    | Accuracy | Precision | Recall | F1-Score |
|----------|----------|----------|-----------|--------|----------|
| Case I   | SVM      | 93.33    | 100       | 87     | 93       |
|          | MLP      | 96.67    | 97        | 97     | 97       |
|          | XGBoost  | 93.97    | 94        | 93     | 89       |
|          | ViT-Head | 80.56    | 44        | 67     | 53       |
|          | TabNet   | 83.33    | 29        | 44     | 59       |
| Case II  | SVM      | 94.17    | 95        | 93     | 94       |
|          | MLP      | 93.83    | 92        | 100    | 96       |
|          | XGBoost  | 90.28    | 78        | 58     | 67       |
|          | ViT-Head | 95.67    | 95        | 93     | 94       |
|          | TabNet   | 89.17    | 94        | 83     | 88       |
| Case III | SVM      | 96.67    | 98        | 95     | 97       |
|          | MLP      | 95.33    | 96        | 95     | 95       |
|          | XGBoost  | 91.11    | 77        | 67     | 71       |
|          | ViT-Head | 93.67    | 94        | 93     | 94       |
|          | TabNet   | 87.78    | 90        | 30     | 45       |
| Case IV  | SVM      | 96.64    | 93        | 97     | 95       |
|          | MLP      | 94.14    | 95        | 93     | 94       |
|          | XGBoost  | 91.46    | 89        | 94     | 92       |
|          | ViT-Head | 90.79    | 89        | 93     | 91       |
|          | TabNet   | 90.62    | 93        | 88     | 90       |

than with Stable Diffusion alone (42/56%). Compared to prior cases, this intense increase, especially in recall—demonstrates that SMOTE feature-level synthesis fills critical gaps not covered by image-level diffusion augmentation, as supported by Saini et al. [74]. The improvement in XGBoost (93% accuracy, 93% recall, 89% F1) and ViT-Head (80.56% accuracy, 67% recall, 53% F1-score) underscores that hybrid augmentation strengthens even diverse model types.

##### 2) CASE II: MODERATE IMBALANCE (CLEAN 300 AND DIRTY 60)

Similarly, in this case, by increasing the sample size as shown in Table 5, DiSMHA consistently improves the dirty panel detection. In this case, SVM achieves 94.17% accuracy and 93% recall, with a 94% F1-score; MLP reaches 95.83% accuracy, 100% recall, and 96% F1-score; TabNet jumps to 83% recall and 88% F1-score, nearly doubling from its earlier baselines. ViT-Head also increased to 93%

**TABLE 6.** Wilcoxon signed rank test results for F1-Score across all cases and models.

| Case                           | Model    | Comparison                 | Statistic | p-value | Significant |
|--------------------------------|----------|----------------------------|-----------|---------|-------------|
| Case I (Clean 150 Dirty 30)    | SVM      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | MLP      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 8.5       | 0.5000  | No          |
|                                | XGBoost  | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | ViT Head | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 3.0       | 0.2500  | No          |
|                                | TabNet   | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
| Case II (Clean 300 Dirty 60)   | SVM      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | MLP      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | XGBoost  | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | ViT Head | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 13.5      | 0.0938  | No          |
|                                | TabNet   | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
| Case III (Clean 750 Dirty 150) | SVM      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | MLP      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | XGBoost  | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | ViT Head | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 14.0      | 0.0625  | No          |
|                                | TabNet   | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 10.0      | 0.0625  | No          |
| Case IV (Clean 1493 Dirty 293) | SVM      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | MLP      | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | XGBoost  | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |
|                                | ViT Head | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 4.5       | 0.3750  | No          |
|                                | TabNet   | DiSMHA vs w/o Augmentation | 15.0      | 0.0312  | Yes         |
|                                |          | DiSMHA vs Stable Diffusion | 15.0      | 0.0312  | Yes         |

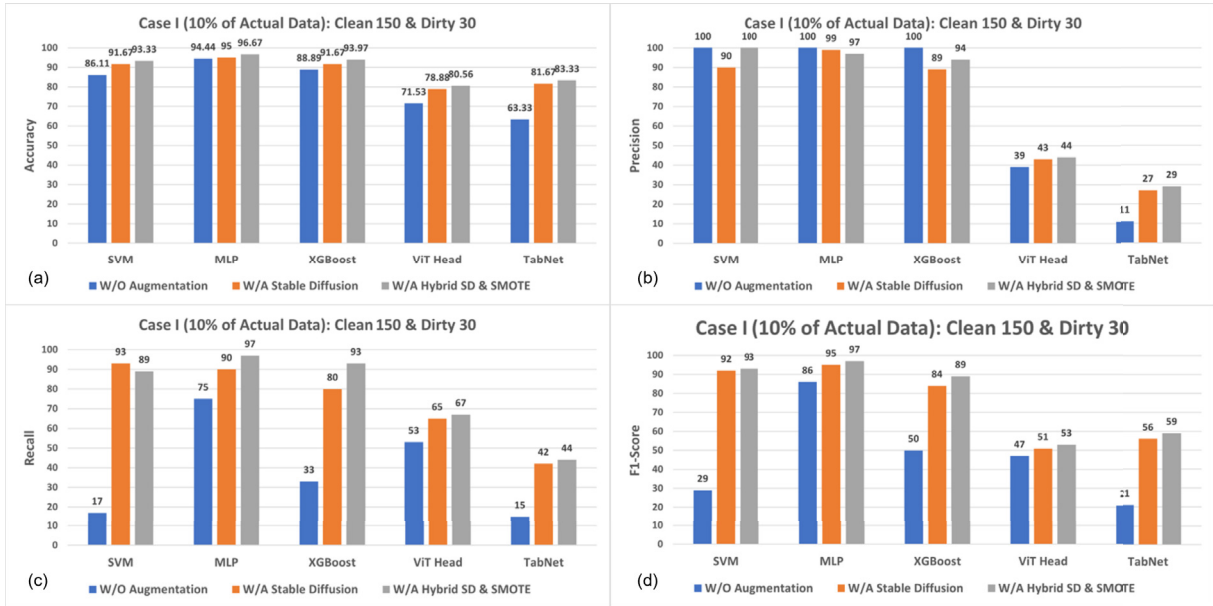
recall and 94% F1-score, highlighting the better performance diversified training data [7]. In all models, recall and F1-score exceed their values from both the unaugmented and Stable Diffusion-only settings, confirming the synergistic effect of pixel- and feature-level balancing.

### 3) CASE III: INCREASED SAMPLE (CLEAN 750 AND DIRTY 150)

In this case III, DiSMHA continues to outperform other augmentation methods, as shown in Table 5. With this increased dataset case, SVM achieves 96.67% accuracy, 95% recall, and a 97% F1-score; MLP follows closely with 95.33% accuracy, 95% recall, and a 95 F1-score. TabNet, previously weakest, now achieves 30% recall and a 45% F1-score, a significant increase compared to its earlier 6/6 baseline and better than 28/41% with Stable Diffusion. XGBoost and ViT-Head both post recall values of 67% and 93%, respectively, and F1-scores of 71% and 94%, indicating that the hybrid approach addresses feature space representation sufficiently as compared to the image-based. This aligns with [18], where combined augmentation led to superior generalization.

### 4) CASE IV: LARGE IMBALANCE (CLEAN 1493 AND DIRTY 293)

With such an increased dataset, in this case, DiSMHA demonstrates the scalability and consistent minority class recovery as shown in Table 5. SVM, MLP, and ViT-Head each achieve recall rates above 93%, with F1-scores of 95%, 94%, and 91%, respectively. TabNet attains 88% recall and a 90% F1-score—remarkably better than its pre-augmentation performance and surpassing its Stable Diffusion-only recall of 88 and F1-score of 89%. XGBoost, though slightly lower, remains robust with 85% recall and a 92% F1-score. The nearly uniform performance across all metrics and models highlights the pipeline's effectiveness in both large and small-scale scenarios. The hybrid DiSMHA approach leverages the complementary strengths of Stable Diffusion (for structural, pixel-level diversity) and SMOTE (for representational, feature-level diversity), thus overcoming the limitations observed with each augmentation alone. This dual augmentation strategy consistently enhances minority class recall and F1-score without improving the overall accuracy or precision, as shown in Fig. 17. Based on the previous research [7], [18], [74], hybrid data augmentation frameworks



**FIGURE 18.** Performance comparison of SVM, MLP, XGBoost, ViT-Head, and TabNet for Case I (150 clean, 30 dirty) across (a) Accuracy, (b) Precision, (c) Recall, and (d) F1-Score, under three settings: without augmentation, with Stable Diffusion, and with DiSMHA (hybrid augmentation).

offer superior mitigation of class imbalance, especially in real-world visual classification and rare event detection scenarios. Our results clearly demonstrate that DiSMHA yields not only higher detection rates for dust-affected solar panels but also balanced model robustness, confirming its suitability for imbalanced, data-scarce applications.

#### E. STATISTICAL ANALYSIS USING WILCOXON TEST

Table 6 presents the results of the Wilcoxon signed rank test performed on the F1-scores for the minority class (dusty) across all experimental cases and classifiers. Across all dataset scenarios, DiSMHA demonstrates statistically significant improvements ( $p < 0.05$ ) over the without Augmentation baseline for every model, affirming the consistent advantage of the proposed hybrid augmentation framework in severely imbalanced settings. Moreover, in most cases and for most models, DiSMHA also achieves statistically significant gains over single augmentation using Stable Diffusion. Notably, significance is consistently observed for classical machine learning models such as SVM, MLP, and XGBoost, while deep architectures like ViT Head and TabNet show diminished or non-significant differences in some larger, less imbalanced datasets. These findings underscore that the impact of DiSMHA is most pronounced where class imbalance is extreme and minority class samples are limited.

#### V. DISCUSSION

The discussion section provides an in-depth analysis of the observed results, synthesizing the impact of the proposed DiSMHA hybrid augmentation approach across different experimental cases (see comparative results in Fig. 18, 19, 20, 21) and classification models. By ranking model

performance, examining the influence of class imbalance, and exploring metric-specific trade-offs, we analyzed the key insights into the effectiveness and generalizability of our pipeline for dust detection in solar panels.

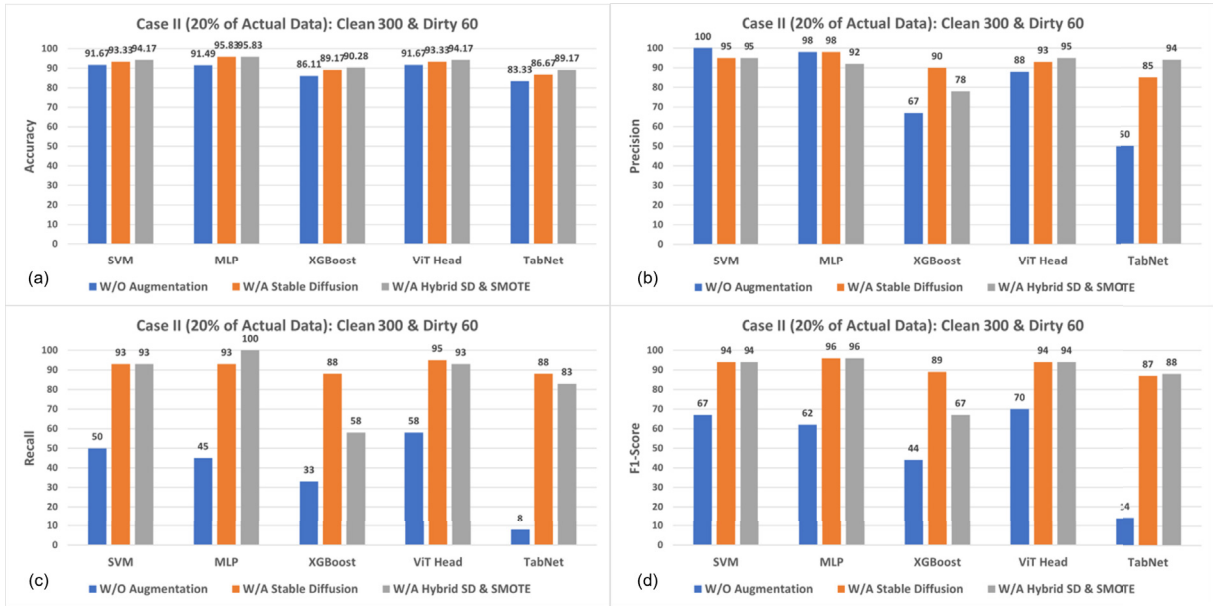
#### A. COMPARATIVE ANALYSIS ACROSS CASES AND MODELS

##### 1) MODEL PERFORMANCE TRENDS

By focusing on the performance observed through classifiers across all cases shown in Fig. 18, 19, 20, 21, MLP consistently emerges as the top performance, achieving F1-scores as high as 97% and recall rates up to 100% in moderately and severely imbalanced cases with DiSMHA. SVM also demonstrates robust and reliable performance, with F1-scores and recall frequently exceeding 93% after hybrid augmentation. Whereas, XGBoost performs well in larger datasets but is less stable in extreme imbalance, while ViT-Head excels in recall as data size increases, as discussed in a recent study on transformer-based architectures [71]. TabNet initially didn't perform well in fewer dataset cases, but in large datasets from hybrid augmentation—its recall and F1-score rise dramatically, especially in the most imbalanced scenarios, highlighting the particular advantage of feature-level oversampling for attention-based tabular models as discussed in Fig. 21. This trend underscores the universal benefit of DiSMHA but especially its value for models that typically underperform in data-scarce, imbalanced contexts.

##### 2) IMPACT OF IMBALANCED RATIO

As from the results, it is observed that the performance improvements with augmentation are most dramatic in case I (shown in Fig. 18) and case II (shown in Fig. 19), where the imbalance is severe. For example, in Case I,



**FIGURE 19.** Performance comparison of SVM, MLP, XGBoost, ViT-Head, and TabNet for Case II (300 clean, 60 dirty) across (a) Accuracy, (b) Precision, (c) Recall, and (d) F1-Score, under three experimental conditions: without augmentation, with Stable Diffusion, and with DiSMHA (hybrid augmentation).

TabNet's recall increases from 21% (no augmentation) to 44% (DiSMHA). At the same time, the F1 score is more than doubles. MLP's recall and F1-score achieve better accuracy results, as discussed in Case II. As the dataset size grows (shown in Fig. 20, 21), gains remain substantial but begin to plateau, indicating diminishing returns—a phenomenon widely reported in class imbalance literature [74]. This suggests that hybrid augmentation is most essential when real minority samples are very less, but once a certain data diversity threshold is crossed, additional synthetic examples yield smaller marginal benefits.

### 3) METRIC-SPECIFIC INSIGHTS

From the obtained performance results, it is observed that precision remains consistently high across all cases, but the performance of the F1 score and recall improved with hybrid augmentation. Without augmentation, models exhibited high precision but low recall—a well-documented consequence of imbalance where the classifier is conservative in predicting the minority class [75]. With DiSMHA, both recall and F1-score improved smartly, resulting in better model performance as shown in Fig. 21. The F1-score, in particular, proves to be the most informative and fair metric in these imbalanced dataset cases, as it captures the interplay between missed detections (recall) and false positives (precision) and effectively summarizes model utility for real-world dust detection and maintenance planning.

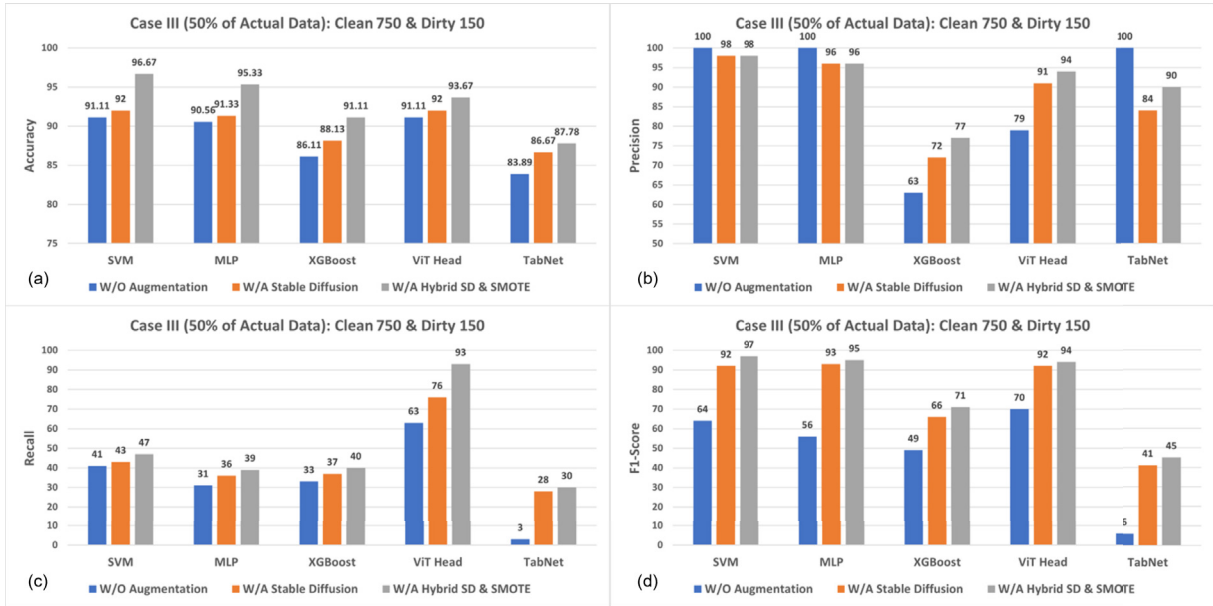
### 4) STATISTICAL SIGNIFICANCE AND ROBUSTNESS

To validate the reliability of our findings, cross-validation was performed across all models and experimental cases

using 5-fold stratified splits, ensuring representative sampling of minority and majority classes. The results consistently demonstrate that augmentation strategies, especially the hybrid DiSMHA approach, significantly enhance performance metrics such as Recall and F1-Score across folds, confirming the robustness of the proposed framework as shown in Fig. 21. Furthermore, the consistency of augmentation benefits across folds highlights the stability of the hybrid augmentation framework under varying data imbalance cases, underscoring its generalizability to real-world imbalanced datasets. This robustness aligns with prior studies demonstrating that advanced augmentation methods can improve minority class recognition while maintaining overall model reliability [7], [18], [76]).

### B. STATISTICAL SIGNIFICANCE OF DiSMHA PERFORMANCE

The Wilcoxon analysis in Table 6 substantiates the robustness and practical value of the DiSMHA framework for real-world imbalanced data problems. The consistent statistical significance across models and cases, especially in the most challenging settings, demonstrates that hybrid augmentation not only boosts overall classification metrics but also provides reliable and repeatable improvement for minority class detection. As the dataset size increases and class distribution becomes more balanced, the added benefit of hybrid augmentation over strong single-method baselines, such as Stable Diffusion, becomes less pronounced, particularly for advanced deep learning models. Nevertheless, the consistent outperformance of DiSMHA over cases without augmentation baseline highlights its potential as an effective solution



**FIGURE 20.** Comparative performance of SVM, MLP, XGBoost, ViT-Head, and TabNet on Case III (750 clean, 150 dirty) across (a) Accuracy, (b) Precision, (c) Recall, and (d) F1-Score, evaluated under three conditions: without augmentation, with Stable Diffusion augmentation, and with DiSMHA hybrid augmentation.

for imbalanced learning scenarios commonly encountered in practice, where capturing minority class patterns is critical.

### C. EVALUATION OF DATA AUGMENTATION TECHNIQUES

This study evaluated two augmentation techniques—Stable Diffusion (SD) based image synthesis and SMOTE feature-level oversampling—individually and in hybrid form (DiSMHA) shown in Fig. 21. Results indicate that Stable Diffusion alone significantly improves minority class recall and F1-score by introducing visually diverse and semantically meaningful synthetic images. SMOTE complements this by enriching the feature space through interpolation of minority samples, thus enhancing class representation [77]. However, neither approach alone achieves the optimal balance between pixel-level and feature-level diversity, which is critical for the robust classification of imbalanced datasets. The hybrid approach outperforms both individual methods, confirming that combining generative image augmentation with feature space oversampling leads to superior model generalization and balanced predictive performance [78].

### D. IMPACT OF HYBRID AUGMENTATION APPROACH

The hybrid DiSMHA method demonstrates consistent and substantial improvements across all evaluation metrics—accuracy, precision, recall, and F1-score—across varying imbalance cases discussed in Fig. 21. Notably, TabNet, a model sensitive to class distribution, exhibits the highest gains with hybrid augmentation, showing a recall increase from 15% (without augmentation) to over 40% as shown in Fig. 21. This underscores DiSMHA's effectiveness in enhancing minority class detection without sacrificing overall

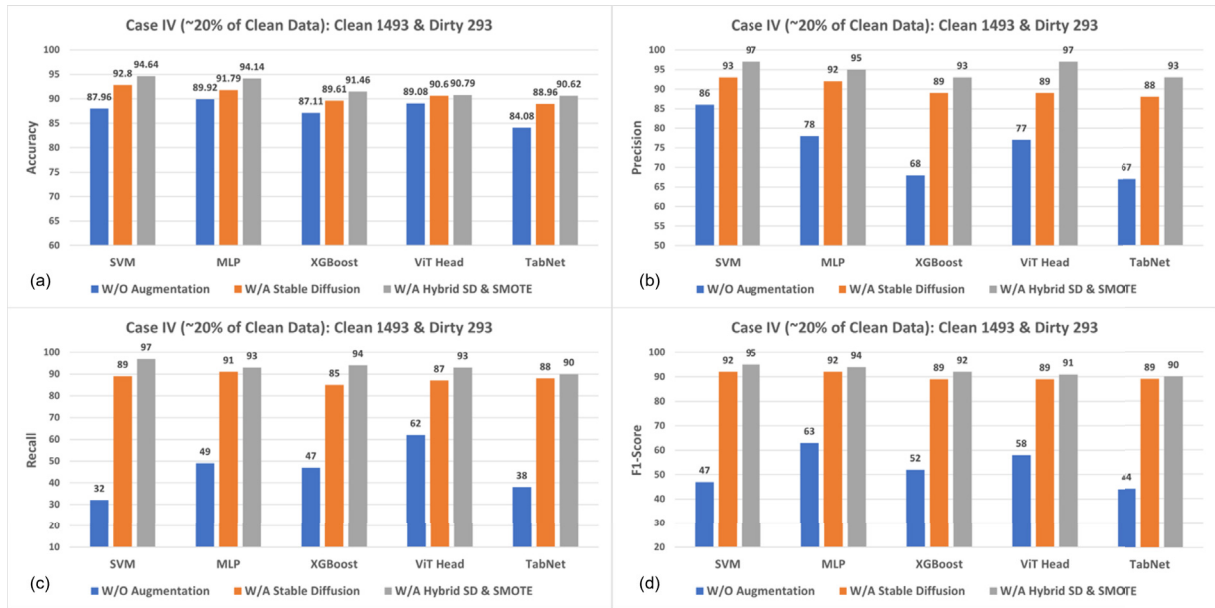
accuracy. The improvement is attributable to the combined effect of realistic synthetic image generation and enriched feature interpolation, addressing limitations of data scarcity and feature homogeneity that traditionally hinder imbalanced learning. These findings are consistent with contemporary literature emphasizing multi-level augmentation for complex visual tasks [79], [80].

### E. STRENGTHS AND LIMITATIONS OF RESEARCH

The key strength of this research is the novel integration of diffusion-based generative models with classical SMOTE in a cohesive pipeline, offering comprehensive data augmentation from pixel to feature space. This hybridization effectively tackles severe imbalance challenges typical in solar panel defect datasets and similar industrial applications. The use of multiple classifiers further validates the generalizability of the approach. Limitations include the computational overhead associated with generative synthesis and feature extraction, which may impede real-time deployment in resource-constrained environments. Additionally, while Stable Diffusion generates diverse synthetic samples, further exploration is needed to automate prompt engineering and optimize synthesis quality [81]. Future work may focus on adaptive augmentation strategies and exploring other generative models to further enhance minority class diversity.

### F. SUITABILITY WITH DIFFERENT DOMAINS

The DiSMHA framework exhibits strong potential across domains characterized by imbalanced visual data. In industrial inspection, where defect samples are small yet critical, hybrid augmentation can enhance detection reliability and



**FIGURE 21.** Comparative performance of SVM, MLP, XGBoost, ViT-Head, and TabNet on Case IV (750 clean, 150 dirty) across (a) Accuracy, (b) Precision, (c) Recall, and (d) F1-Score, evaluated under three conditions: without augmentation, with Stable Diffusion augmentation, and with DiSMHA hybrid augmentation.

**TABLE 7.** Comparison with prior studies and augmentation techniques.

| Ref  | Year | Dataset                                               | Augmentation Technique            | Models                              | Accuracy (%) | Recall (%) | F1-Score (%) |
|------|------|-------------------------------------------------------|-----------------------------------|-------------------------------------|--------------|------------|--------------|
| [83] | 2021 | Pima, WDBC, WPBC, Ionosphere, Breast-cancer-wisconsin | SMOTE                             | SVM, RF, k-NN                       | 85–92        | 78–89      | 80–90        |
| [85] | 2022 | CIFAR-10 (Imbalanced: 99% "cat" class removed)        | Stable Diffusion                  | CNN                                 | 83.5         | 80         | 81           |
| [86] | 2022 | Credit Card Fraud Dataset                             | SMOTE                             | XGBoost, RF                         | 99.9         | 81         | 83           |
| [84] | 2025 | Tabular datasets (e.g., Adult Income)                 | Stable Diffusion (Tab-DDPM)       | DT, GNB, KNN, LR, RF                | 88–91        | 85–89      | 86–90        |
|      | Ours | Solar Panel Dust Dataset                              | Stable Diffusion + SMOTE (Hybrid) | SVM, XGBoost, MLP, ViT-Head, TabNet | 94.4–97      | 97–100.0   | 94–97        |

reduce false negatives. In medical imaging, synthetic sample generation combined with feature-level oversampling can improve diagnosis accuracy for rare pathologies, aligning with studies in histopathology and radiology augmentation [82]. Furthermore, for rare event detection in surveillance or environmental monitoring, the framework's ability to generate high-fidelity synthetic data while preserving feature diversity is invaluable for robust model training. Thus, DiSMHA provides a scalable, domain-agnostic solution to class imbalance, empowering various safety-critical applications.

### G. COMPARISON WITH PRIOR STUDIES

The comparative analysis presented in Table 7 clearly highlights the effectiveness of our proposed hybrid augmentation framework (DiSMHA), integrating both Stable Diffusion and SMOTE, compared to individual augmentation approaches from recent studies. Previous studies primarily using SMOTE alone [83] or Stable Diffusion alone [84]

reported moderate improvements in classification metrics. For instance, Wang et al. [83] applied SMOTE to various tabular datasets such as Pima and Breast Cancer, achieving an accuracy between 85–92%, recall between 78–89%, and F1 scores around 80–90%. Similarly, the Stable Diffusion-based method presented by Hastings Blow et al. [84] on tabular datasets yielded accuracy around 88–91%, recall from 85–89%, and F1 scores ranging between 86–90%. These standalone methods have been effective to some extent, yet they remain limited by either insufficient diversity at the pixel level (SMOTE-only methods) or challenges in feature-space oversampling alone (Stable Diffusion-only methods). In contrast, our DiSMHA method, leveraging a hybrid pixel-level and feature-level augmentation approach, significantly surpasses these standalone augmentation strategies, achieving accuracy, recall, and F1 scores consistently above 94% across multiple models, including SVM, XGBoost, MLP, ViT-Head, and TabNet. Specifically, the recall and F1 score improvements indicate that DiSMHA successfully enhances

the classifier's sensitivity to minority class samples, addressing a critical weakness commonly observed in previous approaches. The superior performance of DiSMHA can be attributed to the integration of Stable Diffusion-based image synthesis, which enriches pixel-level diversity with realistic minority class representations, and SMOTE, which effectively balances the feature space. This dual augmentation provides a richer training dataset, substantially mitigating the common imbalance issues. Furthermore, DiSMHA's robustness is demonstrated by consistently high results across different imbalance scenarios (from severe to moderate imbalance), a strength not explicitly reported in earlier studies.

## VI. CONCLUSION AND FUTURE WORK

### A. SUMMARY OF FINDINGS

This research introduces DiSMHA, a novel hybrid data augmentation framework designed to address the critical challenge of class imbalance in solar panel dust detection tasks. By effectively combining Stable Diffusion-based image synthesis with SMOTE feature-level oversampling, DiSMHA simultaneously enhances both pixel-level diversity and feature-space representation, overcoming key limitations associated with traditional augmentation methods. The approach was rigorously evaluated across four progressively imbalanced dataset cases discussed in Fig. 18, 19, 20, 21, ranging from severe imbalance with a low sample size to more balanced scenarios with larger volumes of data. Experimental results demonstrate consistent and substantial improvements in classification performance metrics, including accuracy, precision, recall, and F1-score, across all cases and multiple state-of-the-art classifiers such as SVM, XGBoost, MLP, ViT-Head, and TabNet. Notably, the hybrid approach yielded marked increases in minority class recall and F1-score, metrics critical for robust detection of dusty solar panels—this improvement addresses the common challenge of minority class neglect in imbalanced datasets. Models like TabNet and MLP exhibited the most significant benefits from the enriched dataset diversity, highlighting DiSMHA's ability to enhance learning in deep and ensemble architectures by providing more representative synthetic samples [87]. The Wilcoxon signed rank test further validated the effectiveness of our proposed DiSMHA framework, revealing statistically significant improvements in minority class detection over baseline methods across all experimental scenarios. This robust statistical evidence underscores the reliability and practical impact of DiSMHA for addressing severe class imbalance. Overall, these findings confirm the framework's potential for real-world applications where accurate recognition of minority classes is critical. In summary, the proposed DiSMHA framework, which integrates Stable Diffusion-based image synthesis and SMOTE feature-level oversampling, demonstrated substantial gains in handling class imbalance for solar panel dust detection. Across four imbalance scenarios, DiSMHA improved minority class

recall by up to 100% and boosted F1-score by 97% compared to non-augmented baselines, with overall model accuracies ranging from 94% to 98%. These improvements highlight the effectiveness of combining pixel-level and feature-level augmentation for robust, domain-agnostic classification.

### B. BROADER IMPLICATIONS AND FUTURE WORK

The proposed DiSMHA framework offers a scalable and domain-agnostic solution for real-world imbalanced classification challenges. Its ability to generate semantically consistent synthetic images and enrich the feature space makes it highly adaptable for industrial inspection, medical diagnosis, and rare event detection where minority class samples are limited yet critical [88]. The demonstrated performance improvements highlight the potential for practical deployment in automated quality control and health monitoring systems. However, challenges remain, particularly in computational efficiency and automated prompt engineering for generative models [89]. Future research will focus on optimizing these aspects to enable real-time augmentation and training workflows, enhancing applicability in dynamic operational environments.

### DECLARATIONS

- **Conflict of interest:** The authors have no conflicts of interest to declare that are relevant to the content of this article.

### ACKNOWLEDGMENT

The authors acknowledge the use of an LLM model for assistance with grammatical corrections and sentence refinements in this manuscript.

### REFERENCES

- [1] M. E. Rayed, S. M. S. Islam, S. I. Niha, J. R. Jim, M. M. Kabir, and M. F. Mridha, "Deep learning for medical image segmentation: State-of-the-art advancements and challenges," *Informat. Med. Unlocked*, vol. 47, Jun. 2024, Art. no. 101504.
- [2] S. Das, S. S. Mullick, and I. Zelinka, "On supervised class-imbalanced learning: An updated perspective and some key challenges," *IEEE Trans. Artif. Intell.*, vol. 3, no. 6, pp. 973–993, Dec. 2022.
- [3] Y. Chen, Y. Ding, F. Zhao, E. Zhang, Z. Wu, and L. Shao, "Surface defect detection methods for industrial products: A review," *Appl. Sci.*, vol. 11, no. 16, p. 7657, Aug. 2021.
- [4] S. Abdallah, M. Sharifa, M. K. I. KH. Almadhoun, M. M. Khawar, U. Shaikh, K. M. Balabel, I. Saleh, A. Manzoor, A. K. Mandal, O. Ekomweren, W. M. Khine, and O. T. Oyelaja, "The impact of artificial intelligence on optimizing diagnosis and treatment plans for rare genetic disorders," *Cureus*, vol. 15, no. 10, pp. 1–12, Oct. 2023.
- [5] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," *J. Big Data*, vol. 6, no. 1, pp. 1–54, Dec. 2019.
- [6] D. Elreedy and A. F. Atiya, "A comprehensive analysis of synthetic minority oversampling technique (SMOTE) for handling class imbalance," *Inf. Sci.*, vol. 505, pp. 32–64, Dec. 2019.
- [7] P. Alimisis, I. Mademlis, P. Radoglou-Grammatikis, P. Sarigiannidis, and G. T. Papadopoulos, "Advances in diffusion models for image data augmentation: A review of methods, models, evaluation metrics and future research directions," *Artif. Intell. Rev.*, vol. 58, no. 4, pp. 1–55, Jan. 2025.
- [8] J. Xie, L. Sun, and Y. F. Zhao, "On the data quality and imbalance in machine learning-based design and manufacturing—A systematic review," *Engineering*, vol. 45, pp. 105–131, Feb. 2025.

- [9] K. Ghosh, C. Bellinger, R. Corizzo, P. Branco, B. Krawczyk, and N. Japkowicz, "The class imbalance problem in deep learning," *Mach. Learn.*, vol. 113, no. 7, pp. 4845–4901, Jul. 2024.
- [10] R. Kumar, G. Baloch, A. Baseer, and J. Bhatti, "Fungal blast disease detection in Rice seed using machine learning," *Int. J. Adv. Comput. Sci. Appl.*, vol. 12, no. 2, pp. 1–18, 2021.
- [11] R. Mohammed, J. Rawashdeh, and M. Abdullah, "Machine learning with oversampling and undersampling techniques: Overview study and experimental results," in *Proc. 11th Int. Conf. Inf. Commun. Syst. (ICICS)*, Apr. 2020, pp. 243–248.
- [12] T. Wongvorachan, S. He, and O. Bulut, "A comparison of undersampling, oversampling, and SMOTE methods for dealing with imbalanced classification in educational data mining," *Information*, vol. 14, no. 1, p. 54, Jan. 2023.
- [13] G. Rekha, A. K. Tyagi, and V. K. Reddy, "Performance analysis of under-sampling and over-sampling techniques for solving class imbalance problem," *SSRN Electron. J.*, pp. 1305–1315, Feb. 2019.
- [14] H. Ding, B. Wei, Z. Gu, Z. Yu, H. Zheng, B. Zheng, and J. Li, "KA-ensemble: Towards imbalanced image classification ensembling under-sampling and over-sampling," *Multimedia Tools Appl.*, vol. 79, nos. 21–22, pp. 14871–14888, Jun. 2020.
- [15] J. Zhang and L. Chen, "Clustering-based undersampling with random over sampling examples and support vector machine for imbalanced classification of breast cancer diagnosis," *Comput. Assist. Surgery*, vol. 24, no. sup2, pp. 62–72, Oct. 2019.
- [16] I. Dey and V. Pratap, "A comparative study of SMOTE, borderline-SMOTE, and ADASYN oversampling techniques using different classifiers," in *Proc. 3rd Int. Conf. Smart Data Intell. (ICSMDI)*, Mar. 2023, pp. 294–302.
- [17] A. F. N. Masruriyah, H. Y. Novita, C. E. Sukmawati, A. Fauzi, D. Wahiddin, and H. H. Handayani, "Thorough evaluation of the effectiveness of SMOTE and ADASYN oversampling methods in enhancing supervised learning performance for imbalanced heart disease datasets," in *Proc. 8th Int. Conf. Informat. Comput. (ICIC)*, Dec. 2023, pp. 1–7.
- [18] C.-Y. Lin, T.-L. Tseng, S. H. Emon, and T.-H. Tsai, "Generative AI-driven data augmentation for robust virtual metrology: GANs, VAEs, and diffusion models," *IEEE Trans. Semicond. Manuf.*, vol. 38, no. 3, pp. 642–658, Aug. 2025.
- [19] A. Ayton and D. Nelson, "Diffusion models as data augmentation for medical imaging," 2020, *arXiv:2310.16794v2*.
- [20] K. Rais, M. Amroune, and M. Y. Haouam, "Medical image generation techniques for data augmentation: Disc-VAE versus GAN," in *Proc. 6th Int. Conf. Pattern Anal. Intell. Syst. (PAIS)*, Apr. 2024, pp. 1–8.
- [21] A. Suresh and C. D'Ettore, "Revolutionizing medical imaging: The vital role of diffusion models in modern image augmentation," *Lens J.*, vol. 2, no. 2, pp. 9–23, 2024, doi: [10.5281/zenodo.11198187](https://doi.org/10.5281/zenodo.11198187).
- [22] R. Raj, J. Mathew, S. K. Kannath, and J. Rajan, "Crossover based technique for data augmentation," *Comput. Methods Programs Biomed.*, vol. 218, May 2022, Art. no. 106716.
- [23] W. Shafqat and Y.-C. Byun, "A hybrid GAN-based approach to solve imbalanced data problem in recommendation systems," *IEEE Access*, vol. 10, pp. 11036–11047, 2022.
- [24] H. Ding, L. Chen, L. Dong, Z. Fu, and X. Cui, "Imbalanced data classification: A KNN and generative adversarial networks-based hybrid approach for intrusion detection," *Future Gener. Comput. Syst.*, vol. 131, pp. 240–254, Jun. 2022.
- [25] S. Vivekananthan, "Comparative analysis of generative models: Enhancing image synthesis with VAEs, GANs, and stable diffusion," 2024, *arXiv:2408.08751*.
- [26] P. Kumar, "Diffusion models and generative artificial intelligence: Frameworks, applications and challenges," *Arch. Comput. Methods Eng.*, vol. 2025, pp. 1–44, Apr. 2025.
- [27] J. Kanyepe, R. Boy, M. Chibaro, T. Mphela, and K. Tlhakanelo, "Revolutionizing supply chain forecasting with generative AI and machine learning," in *Supply Chain Transformation Through Generative AI and Machine Learning*. Hershey, PA, USA: IGI Global Scientific Publishing, 2025, pp. 435–462.
- [28] E. Strelcenia and S. Prakoornwit, "A survey on GAN techniques for data augmentation to address the imbalanced data issues in credit card fraud detection," *Mach. Learn. Knowl. Extraction*, vol. 5, no. 1, pp. 304–329, Mar. 2023.
- [29] M. Temraz and M. T. Keane, "Solving the class imbalance problem using a counterfactual method for data augmentation," *Mach. Learn. Appl.*, vol. 9, Sep. 2022, Art. no. 100375.
- [30] A. S. Dina, A. B. Siddique, and D. Manivannan, "Effect of balancing data using synthetic data on the performance of machine learning classifiers for intrusion detection in computer networks," *IEEE Access*, vol. 10, pp. 96731–96747, 2022.
- [31] Y. Nie, A. S. Zamzam, and A. Brandt, "Resampling and data augmentation for short-term PV output prediction based on an imbalanced sky images dataset using convolutional neural networks," *Sol. Energy*, vol. 224, pp. 341–354, Aug. 2021.
- [32] X. Chao and L. Zhang, "Few-shot imbalanced classification based on data augmentation," *Multimedia Syst.*, vol. 29, no. 5, pp. 2843–2851, Oct. 2023.
- [33] Z. Allal, H. N. Noura, O. Salman, and K. Chahine, "Leveraging the power of machine learning and data balancing techniques to evaluate stability in smart grids," *Eng. Appl. Artif. Intell.*, vol. 133, Jul. 2024, Art. no. 108304.
- [34] H. Kaur, H. S. Pannu, and A. K. Malhi, "A systematic review on imbalanced data challenges in machine learning: Applications and solutions," *ACM Comput. Surveys*, vol. 52, no. 4, pp. 1–36, Jul. 2020.
- [35] F. Thabtah, S. Hammoud, F. Kamalov, and A. Gonsalves, "Data imbalance in classification: Experimental evaluation," *Inf. Sci.*, vol. 513, pp. 429–441, Mar. 2020.
- [36] G. Sambasivam and G. D. Opiyo, "A predictive machine learning application in agriculture: Cassava disease detection and classification with imbalanced dataset using convolutional neural networks," *Egyptian Informat. J.*, vol. 22, no. 1, pp. 27–34, Mar. 2021.
- [37] I. D. Mienye and Y. Sun, "Performance analysis of cost-sensitive learning methods with application to imbalanced medical data," *Informat. Med. Unlocked*, vol. 25, Jul. 2021, Art. no. 100690.
- [38] J. Zhang, M. Zheng, J. Nan, H. Hu, and N. Yu, "A novel evaluation metric for deep learning-based side channel analysis and its extended application to imbalanced data," *IACR Trans. Cryptograph. Hardw. Embedded Syst.*, vol. 2020, no. 3, pp. 73–96, Jun. 2020.
- [39] H. S. Garladinne, "Solar panel dust detection, [dataset]," Kaggle Inc., San Francisco, CA, USA, 2023. Accessed: Apr. 23, 2025. [Online]. Available: <https://www.kaggle.com/datasets/hemanthsai7/solar-panel-dust-detection>
- [40] S. Hossain, A. M. Arika, I. N. Fahim, J. Uddin, A. Ahmed, H. J. Apon, and M. A. Hoque, "Enhancing solar panel performance: A machine learning approach to dust detection and automated water sprinkle-based cleaning strategy," *Sol. Energy*, vol. 287, Feb. 2025, Art. no. 113240.
- [41] F. Barrientos-Espillo, G. Pajares, J. A. López-Orozco, and E. Besada-Portas, "Customization of the text-to-image diffusion model by fine-tuning for the generation of synthetic images of cyanobacterial blooms in lentic water bodies," *Expert Syst. Appl.*, vol. 287, Aug. 2025, Art. no. 128169.
- [42] A. O'Malley, M. Veenhuizen, and A. Ahmed, "Ensuring appropriate representation in artificial intelligence-generated medical imagery: Protocol for a methodological approach to address skin tone bias," *JMIR AI*, vol. 3, Nov. 2024, Art. no. e58275.
- [43] P. Ashwini and S. H. Bharathi, "Continuous feature learning representation to XGBoost classifier on the aggregation of discriminative features using DenseNet-121 architecture and ResNet 18 architectures towards apraxia recognition in the child speech therapy," *Int. J. Speech Technol.*, vol. 27, no. 1, pp. 187–199, Mar. 2024.
- [44] B. Yousaf, A. F. Mirza, M. Irfan, M. Mansoor, and Z. Yang, "Flarenet: A feature fusion based method for fire detection under diverse conditions," Research Square LLC, Rockville, MD, USA, Tech. Rep., 2024. [Online]. Available: <https://doi.org/10.21203/rs.3.rs-5017887/v1>
- [45] S. Pichandi, G. Balasubramanian, and V. Chakrapani, "Hybrid deep models for parallel feature extraction and enhanced emotion state classification," *Sci. Rep.*, vol. 14, no. 1, p. 24957, Oct. 2024.
- [46] B. Schölkopf and A. J. Smola, "Support vector machines," in *Machine Learning*. Amsterdam, The Netherlands: Elsevier, 2020, pp. 101–121.
- [47] A. Tharwat, "Parameter investigation of support vector machine classifier with kernel functions," *Knowl. Inf. Syst.*, vol. 61, no. 3, pp. 1269–1302, Dec. 2019.
- [48] J. Gu, L. Wang, H. Wang, and S. Wang, "A novel approach to intrusion detection using SVM ensemble with feature augmentation," *Comput. Secur.*, vol. 86, pp. 53–62, Sep. 2019.
- [49] A. Wilson and M. R. Anwar, "The future of adaptive machine learning algorithms in high-dimensional data processing," *Int. Trans. Artif. Intell. (ITALIC)*, vol. 3, no. 1, pp. 97–107, Nov. 2024.

- [50] M. Tahir, B. Yude, T. Mehmood, S. Bashir, Y. Zhenping, and M. Awais, "Classification of LAMOST spectra of B-type and hot subdwarf stars using kernel support vector machine," *Sci. Rep.*, vol. 14, no. 1, p. 16815, Jul. 2024.
- [51] E. Alpaydin, "Multilayer perceptrons," in *Computational Intelligence: A Methodological Introduction*. Cham, Switzerland: Springer, 2010, pp. 233–277.
- [52] T. Georgiou, Y. Liu, W. Chen, and M. Lew, "A survey of traditional and deep learning-based feature descriptors for high dimensional data in computer vision," *Int. J. Multimedia Inf. Retr.*, vol. 9, no. 3, pp. 135–170, Sep. 2020.
- [53] S. Sharma and A. Gosain, "Addressing class imbalance in remote sensing using deep learning approaches: A systematic literature review," *Evol. Intell.*, vol. 18, no. 1, pp. 1–28, Feb. 2025.
- [54] S. Demir and E. K. Sahin, "An investigation of feature selection methods for soil liquefaction prediction based on tree-based ensemble algorithms using AdaBoost, gradient boosting, and XGBoost," *Neural Comput. Appl.*, vol. 35, no. 4, pp. 3173–3190, Feb. 2023.
- [55] P. Zhang, Y. Jia, and Y. Shang, "Research and application of XGBoost in imbalanced data," *Int. J. Distrib. Sensor Netw.*, vol. 18, no. 6, Jun. 2022, Art. no. 155013292211069.
- [56] M. Imani, A. Beikmohammadi, and H. R. Arabnia, "Comprehensive analysis of random forest and XGBoost performance with SMOTE, ADASYN, and GNUS under varying imbalance levels," *Technologies*, vol. 13, no. 3, p. 88, Feb. 2025.
- [57] B. Palanisamy, V. Hassija, A. Chatterjee, A. Mandal, D. Chakraborty, A. Pandey, G. S. S. Chalapathi, and D. Kumar, "Transformers for vision: A survey on innovative methods for computer vision," *IEEE Access*, vol. 13, pp. 95496–95523, 2025.
- [58] M. Heidari, R. Azad, S. Ghorbani Kolahi, R. Arimond, L. Nigge-meier, A. Sulaiman, A. Bozorgpour, E. Khodapanah Aghdam, A. Kazerouni, I. Hachililoglu, and D. Merhof, "Enhancing efficiency in vision transformer networks: Design techniques and insights," 2024, *arXiv:2403.19882*.
- [59] J. Maurício, I. Domingues, and J. Bernardino, "Comparing vision transformers and convolutional neural networks for image classification: A literature review," *Appl. Sci.*, vol. 13, no. 9, p. 5521, Apr. 2023.
- [60] I. Kar, S. Mukhopadhyay, and Z. Ralte, "Enhanced solar panel multiclass fine grained classification using attention distilled self-supervised learning and combined loss," in *Proc. 10th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, Mar. 2024, pp. 133–139.
- [61] M. N. H. Chowdhury, M. Bin Ibne Reaz, S. H. M. Ali, M. L. Crespo, S. Ahmad, G. M. Salim, F. Haque, L. G. G. Ordóñez, M. J. Islam, T. M. Mahdee, K. S. Zaman, M. S. K. Hemel, and M. A. S. Bhuiyan, "Deep learning for early detection of chronic kidney disease stages in diabetes patients: A TabNet approach," *Artif. Intell. Med.*, vol. 166, Aug. 2025, Art. no. 103153.
- [62] A. Roy, M. Saffar, A. Vaswani, and D. Grangier, "Efficient content-based sparse attention with routing transformers," *Trans. Assoc. for Comput. Linguistics*, vol. 9, pp. 53–68, Feb. 2021.
- [63] S. R. Mohanty, M. U. Maruf, V. Singh, and Z. Ahmad, "Machine learning approaches for automatic defect detection in photovoltaic systems," 2024, *arXiv:2409.16069*.
- [64] B. Awuku, Y. Huang, N. Yodo, and E. Asa, "Interpretable machine learning models for failure cause prediction in imbalanced oil pipeline data," *Meas. Sci. Technol.*, vol. 35, no. 7, Jul. 2024, Art. no. 076006.
- [65] R. Yacoubi and D. Axman, "Probabilistic extension of precision, recall, and F1 score for more thorough evaluation of classification models," in *Proc. 1st Workshop Eval. Comparison NLP Syst.*, 2020, pp. 79–91.
- [66] R. A. Poldrack, G. Huckins, and G. Varoquaux, "Establishment of best practices for evidence for prediction: A review," *JAMA psychiatry*, vol. 77, no. 5, pp. 534–540, 2020.
- [67] W. Cullerne Bown, "Sensitivity and specificity versus precision and recall, and related dilemmas," *J. Classification*, vol. 41, no. 2, pp. 402–426, Jul. 2024.
- [68] R. Diallo, C. Edalo, and O. O. Awe, "Machine learning evaluation of imbalanced health data: A comparative analysis of balanced accuracy, MCC, and F1 score," in *Practical Statistical Learning and Data Science Methods: Case Studies From LISA 2020 Global Network, USA*. Cham, Switzerland: Springer, 2024, pp. 283–312.
- [69] B. H. Aubaidan, R. A. Kadir, M. T. Lajb, M. Anwar, K. N. Qureshi, B. A. Taha, and K. Ghafoor, "A review of intelligent data analysis: Machine learning approaches for addressing class imbalance in healthcare-challenges and perspectives," *Intell. Data Anal.*, vol. 29, no. 3, pp. 699–719, 2024.
- [70] E. Jafarigol and T. Trafalis, "A review of machine learning techniques in imbalanced data and future trends," 2023, *arXiv:2310.07917*.
- [71] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
- [72] X. L. Lan, "Traditional augmentation versus deep generative diffusion augmentation for addressing class imbalance in chest X-ray classification," M.S. thesis, Dept. Inf. Comput. Sci., Utrecht Univ., Utrecht, The Netherlands, 2023.
- [73] A. Makhoulouf, M. Maayah, N. Abughanani, and C. Catal, "The use of generative adversarial networks in medical image augmentation," *Neural Comput. Appl.*, vol. 35, no. 34, pp. 24055–24068, Dec. 2023.
- [74] M. Saini and S. Susan, "Tackling class imbalance in computer vision: A contemporary review," *Artif. Intell. Rev.*, vol. 56, no. S1, pp. 1279–1335, Oct. 2023.
- [75] W. D. G. de Oliveira and L. Berton, "A systematic review for class-imbalance in semi-supervised learning," *Artif. Intell. Rev.*, vol. 56, no. S2, pp. 2349–2382, Nov. 2023.
- [76] M. Fahmy Amin, "Confusion matrix in binary classification problems: A step-by-step tutorial," *J. Eng. Res.*, vol. 6, no. 5, pp. 1–16, Dec. 2022.
- [77] S. Maldonado, C. Vairetti, A. Fernandez, and F. Herrera, "FW-SMOTE: A feature-weighted oversampling approach for imbalanced classification," *Pattern Recognit.*, vol. 124, Apr. 2022, Art. no. 108511.
- [78] F. Gurcan and A. Soylu, "Synthetic boosted resampling using deep generative adversarial networks: A novel approach to improve cancer prediction from imbalanced datasets," *Cancers*, vol. 16, no. 23, p. 4046, Dec. 2024.
- [79] C. Shyalika, R. Wickramarachchi, F. El Kalach, R. Harik, and A. Sheth, "Evaluating the role of data enrichment approaches towards rare event analysis in manufacturing," *Sensors*, vol. 24, no. 15, p. 5009, Aug. 2024.
- [80] C. S. J. Kankanamalage, R. Wickramarachchi, F. El Kalach, R. Harik, and A. Sheth, "Evaluating the role of data enrichment approaches towards rare event analysis in manufacturing," *Available SSRN*, Feb. 2024.
- [81] W. Li, J. Wang, and X. Zhang, "PROMPTIST: Automated prompt optimization for text-to-image synthesis," in *Proc. CCF Int. Conf. Natural Lang. Process. Chin. Comput.*, 2024, pp. 295–306.
- [82] T. Islam, M. S. Hafiz, J. R. Jim, M. M. Kabir, and M. F. Mridha, "A systematic review of deep learning data augmentation in medical imaging: Recent advances and future research directions," *Healthcare Analytics*, vol. 5, Jun. 2024, Art. no. 100340.
- [83] S. Wang, Y. Dai, J. Shen, and J. Xuan, "Research on expansion and classification of imbalanced data based on SMOTE algorithm," *Sci. Rep.*, vol. 11, no. 1, p. 24039, Dec. 2021.
- [84] C. Hastings Blow, L. Qian, C. Gibson, P. Obiomon, and X. Dong, "Data augmentation via diffusion model to enhance AI fairness," *Frontiers Artif. Intell.*, vol. 8, Mar. 2025, Art. no. 1530397.
- [85] T. Ø. Eliassen and Y. Ma, "Data synthesis with stable diffusion for dataset imbalance-computer vision," Stanford Univ., Stanford, CA, USA, Tech. Rep. CS230-2022-17, 2022.
- [86] A. Kummer, T. Ruppert, T. Medvegy, and J. Abonyi, "Machine learning-based software sensors for machine state monitoring—The role of SMOTE-based data augmentation," *Results Eng.*, vol. 16, Dec. 2022, Art. no. 100778.
- [87] Y. Li, Z. Shi, C. Liu, W. Tian, Z. Kong, and C. B. Williams, "Augmented time regularized generative adversarial network (ATR-GAN) for data augmentation in online process anomaly detection," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 4, pp. 3338–3355, Oct. 2022.
- [88] S. Fayaz, S. Z. A. Shah, N. M. U. Din, N. Gul, and A. Assad, "Advancements in data augmentation and transfer learning: A comprehensive survey to address data scarcity challenges," *Recent Adv. Comput. Sci. Commun.*, vol. 17, no. 8, pp. 14–35, Nov. 2024.
- [89] P. Korzynski, G. Mazurek, P. Krzykowska, and A. Kurasinski, "Artificial intelligence prompt engineering as a new digital competence: Analysis of generative AI technologies such as ChatGPT," *Entrepreneurial Bus. Econ. Rev.*, vol. 11, no. 3, pp. 25–37, 2023.



**RAJ KUMAR** received the B.E. degree in electrical engineering with a specialization in electronics from Sukkur IBA University, Pakistan. He is currently pursuing the M.S. degree in electronics engineering with Jeju National University, South Korea. He was a Research Engineer (Remote) in AI and robotics at Augmented Startups, South Africa. He was a Machine Learning Engineer at MasterMinds Enterprises, Karachi, Pakistan. He has experience as an Electrical Engineer Supervisor at AGM Government Contractor, Hyderabad, Pakistan. He is also a Research Scholar with the Machine Learning Laboratory, Jeju National University. His research interests include machine learning, robotics, and computer vision.



**YUNG-CHEOL BYUN** received the B.S. degree from Jeju National University, in 1993, and the M.S. and Ph.D. degrees from Yonsei University, in 1995 and 2001, respectively. He was a Special Lecturer with Samsung Electronics and SDS, from 1998 to 2001. From 2001 to 2003, he was a Senior Researcher with the Electronics and Telecommunications Research Institute (ETRI). He was promoted to join Jeju National University, as an Assistant Professor, in 2003, where he is currently a Full Professor with the Computer Engineering Department. His research interests include the areas of AI and machine learning, pattern recognition, blockchain and deep learning-based applications, big data and knowledge discovery, time-series data analysis and prediction, image processing, medical image applications, and recommendation systems.

...



**YONG-WOON KIM** received the B.S. and M.S. degrees in computer science from Yonsei University, South Korea, in 1994 and 1997, respectively, and the Ph.D. degree in computer science and engineering from CHRIST University, Bengaluru, India, in 2023. He is currently a Visiting Professor with Jeju National University, South Korea. Before joining Jeju National University, he was with Samsung Electronics Company Ltd., LG Electronics Company Ltd., and many global IT companies, as a Principal Embedded Software Engineer for ten years, and has been an entrepreneur in India since 2007. He was also an Associate Professor and a Coordinator with the Centre for Digital Innovation, CHRIST University. He completed research collaborations with the Electronics and Telecommunications Research Institute (ETRI) and other global research and development organizations. His research interests include computer vision, artificial intelligence, green hydrogen, and the Internet of Things.